Gravitational Memory Test-Code

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All of the code in this document is purely for testing and development. The final code used in my thesis can be found at https://github.com/Simont12321/Gravitational-memory-thesis.

To Do

- Redo +polarization in 3D with rotational about the correct axis of by rotating the polarization by ω.
- Do a cross polarized wave in 3D.

Variables and Functions and testing

In[ ]:=
```
ClearAll["Global`*"]
```

Rotation matrices applied to GW polarization tensors for a GW traveling along the z-axis.

In[ ]:=
```
Rz[ϕ_] := {{Cos[ϕ], -Sin[ϕ], 0}, {Sin[ϕ], Cos[ϕ], 0}, {0, 0, 1}} (*z-axis rotation*)
```

In[ ]:=
```
P = {{1, 0, 0}, {0, -1, 0}, {0, 0, 0}}; (*+ polarized wave*)
G = {{p, x, 0}, {x, -p, 0}, {0, 0, 0}}; (*general wave polarization*)
X = {{0, 1, 0}, {1, 0, 0}, {0, 0, 0}}; (*cross polarized wave*)
```

Applying the rotation matrix to + polarization. Tensors rotate as T' = RTR^{-1}
To get the new tensor you apply the rotation matrix R and its inverse R^{-1}

```
Simplify[Rz[ϕ].P.Transpose[Rz[ϕ]]] // MatrixForm
```

Out[ ]//MatrixForm=

$$\begin{pmatrix} \text{Cos}[2\,\phi] & \text{Sin}[2\,\phi] & 0 \\ \text{Sin}[2\,\phi] & -\text{Cos}[2\,\phi] & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

Rotations about the y and z-axes

In[ ]:=
```
Ry[α_] := {{Cos[α], 0, Sin[α]}, {0, 1, 0}, {-Sin[α], 0, Cos[α]}}
(*y-axis rotation*)
Rx[α_] := {{1, 0, 0}, {0, Cos[α], -Sin[α]}, {0, Sin[α], Cos[α]}}
 (*x-axis rotation*)
```

Applying rotations:

Simplest case: Plus polarized wave rotated -α degrees about the x-axis and then ϕ degrees about the z-axis.

In[∘]:=
```
Simplify[Rz[ϕ].Rx[-α].P.Transpose[Rx[-α]].Transpose[Ry[ϕ]]] // MatrixForm
```

Out[∘]//MatrixForm=

$$\begin{pmatrix} \text{Cos}[\phi]^2 - \text{Cos}[\alpha]\,\text{Sin}[\alpha]\,\text{Sin}[\phi]^2 & \text{Cos}[\alpha]^2\,\text{Sin}[\phi] & -\text{Cos}[\phi]\,(1+\text{Cos}[\alpha]\,\text{Sin}[\alpha])\,\text{Sin}[\phi] \\ \text{Cos}[\phi]\,(1+\text{Cos}[\alpha]\,\text{Sin}[\alpha])\,\text{Sin}[\phi] & -\text{Cos}[\alpha]^2\,\text{Cos}[\phi] & \text{Cos}[\alpha]\,\text{Cos}[\phi]^2\,\text{Sin}[\alpha] - \text{Sin}[\phi]^2 \\ -\text{Sin}[\alpha]^2\,\text{Sin}[\phi] & \text{Cos}[\alpha]\,\text{Sin}[\alpha] & -\text{Cos}[\phi]\,\text{Sin}[\alpha]^2 \end{pmatrix}$$

The most complex case: GW with a general plus polarization p and cross polarisation x rotated α degrees about the x-axis and then ϕ degrees about the z-axis.

In[∘]:=
```
Simplify[Rz[ϕ].Rx[-α].G.Transpose[Rx[-α]].Transpose[Ry[ϕ]]] // MatrixForm
```

Out[∘]//MatrixForm=

$$\begin{pmatrix} \text{p}\,\text{Cos}[\phi]^2 - \text{x}\,\text{Cos}[\phi]\,(\text{Cos}[\alpha]+\text{Sin}[\alpha])\,\text{Sin}[\phi] - \text{p}\,\text{Cos}[\alpha]\,\text{Sin}[\alpha]\,\text{Sin}[\phi]^2 & \text{Cos}[\alpha]\,(\text{x} \\ \text{Cos}[\alpha]\,\text{Cos}[\phi]\,(\text{x}\,\text{Cos}[\phi]+\text{p}\,\text{Sin}[\alpha]\,\text{Sin}[\phi]) + \text{Sin}[\phi]\,(\text{p}\,\text{Cos}[\phi]-\text{x}\,\text{Sin}[\alpha]\,\text{Sin}[\phi]) & \text{Cos}[\alpha]\,(-\text{p} \\ -\text{Sin}[\alpha]\,(\text{x}\,\text{Cos}[\phi]+\text{p}\,\text{Sin}[\alpha]\,\text{Sin}[\phi]) & \end{pmatrix}$$

Other variables

Constants:

c=1,

dG Earth to GW source distance,

tP time photon is received on Earth,

tG time GW is received on Earth which we set to 0,

(ϕ,θ) location on Earth's sky where the photon is received

h0 GW strain amplitude measured on Earth

t time as measured on Earth

Functions:

Functions are capitalized. A is the function that gives α for given parameters, RP is the function that gives rP.

rP distance from GW source to photon

α angle between GW source and photon (measured up from GW propegation vector)

dP distance from Earth to the photon

tmin initial time the photon encounters the GW

n vector pointing to point on the sky where the photon is found

In[ ]:=

```
$Assumptions = dG > 0 && h0 > 0 && tP > tG && 0 ≤ θ ≤ 2 π && t ∈ Reals;
c = 1;
tG = 0;
M[dP_, dG_] := If[dP > dG, ArcCos[dG/dP], 0];
(*A[dP_, dG_, θ_] :=  If[θ≤M[dP, dG],ArcTan[(dP Sin[θ])/(dG - dP Cos[θ])]+π,
  If[θ≤2π-M[dP, dG],ArcTan[(dP Sin[θ])/(dG - dP Cos[θ])],ArcTan[(dP Sin[θ])/(dG - dP Cos[θ])]-π]] *)
DP[tP_, t_] := c (tP - t)
(*DP[tP_, t_] := If[tP>t,c(tP-t), 0]*)
(*solves for if tP > t, but shouldn't be needed,
although it did change the ϵ and h expresions a little*)
Tmin[tG_, tP_, dG_, θ_] := (-c tG^2 + 2 dG tG + c tP^2 - 2 dG tP Cos[θ])/(-2 c tG + 2 dG + 2 c tP - 2 dG Cos[θ])
RP[dG_, tP_, t_, θ_] := Sqrt[dG^2 + (tP - t)^2 - 2 dG (tP - t) Cos[θ] ]
TrigSimp[opp_, adj_] := FullSimplify[2 * opp/Sqrt[adj^2 + opp^2] * adj/Sqrt[adj^2 + opp^2]]
 (*Simplifies Sin[2ArcTan[opp/adj]]*)(*Sin(2x) = 2CosxSinx*)
```

In[ ]:=

```
Tmin[tG, tP, dG, θ]
```

Out[ ]=

$$\frac{tP^2 - 2\, dG\, tP \cos[\theta]}{2\, dG + 2\, tP - 2\, dG \cos[\theta]}$$

In[ ]:=

```
RP[dG, tP, t , θ]
```

Out[ ]=

$$\sqrt{dG^2 + (-t + tP)^2 - 2\, dG\, (-t + tP) \cos[\theta]}$$

In[ ]:=

```
Simplify[RP[dG, tP, Tmin[tG, tP, dG, θ] , θ] - Tmin[tG, tP, dG, θ] - dG]
```

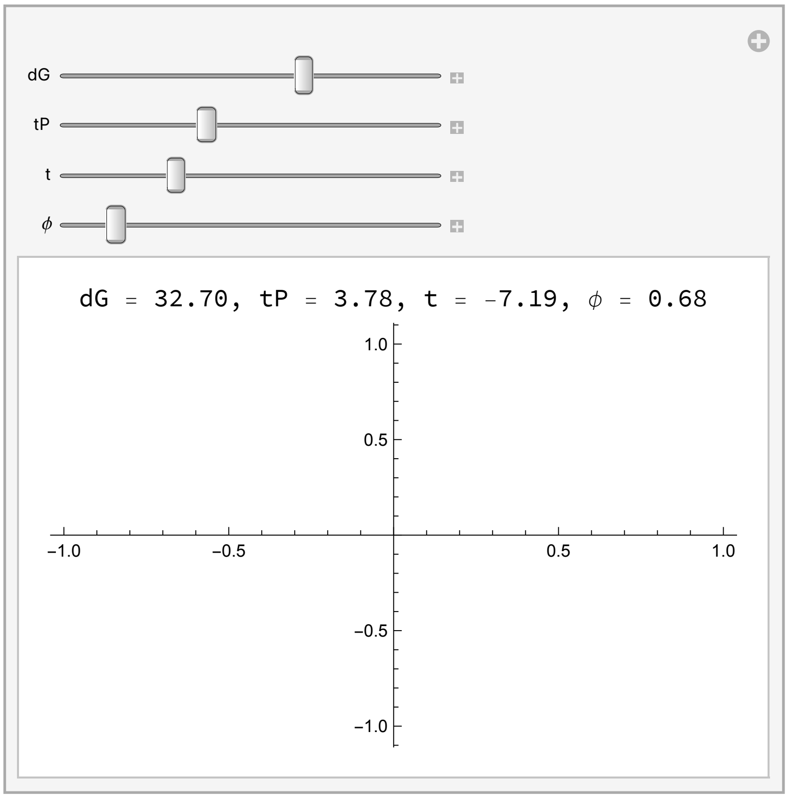
Out[ ]=

0

In[ ]:=
```
A[dP_, dG_, θ_, ϕ_] := If[θ ≤ M[dP, dG], ArcTan[(dP Sin[θ] Cos[ϕ])/(dG - dP Cos[θ])] + π,
  If[θ ≤ 2 π - M[dP, dG], ArcTan[(dP Sin[θ] Cos[ϕ])/(dG - dP Cos[θ])], ArcTan[(dP Sin[θ] Cos[ϕ])/(dG - dP Cos[θ])] - π]]
```

In[ ]:=
```
Manipulate[Labeled[Plot[A[DP[tP, t] , dG, θ, ϕ], {θ, 0, 2 π}, PlotRange → All],
  Row[{"dG = ", NumberForm[dG, {3, 2}], ", tP = ", NumberForm[tP, {3, 2}],
    ", t = ", NumberForm[t, {3, 2}], ", ϕ = ", NumberForm[ϕ, {3, 2}]}], Top],
 {dG, 0.1, 50}, {tP, 0.1, 10}, {t, -10, -0.1}, {ϕ, 0, 2 π}]
```

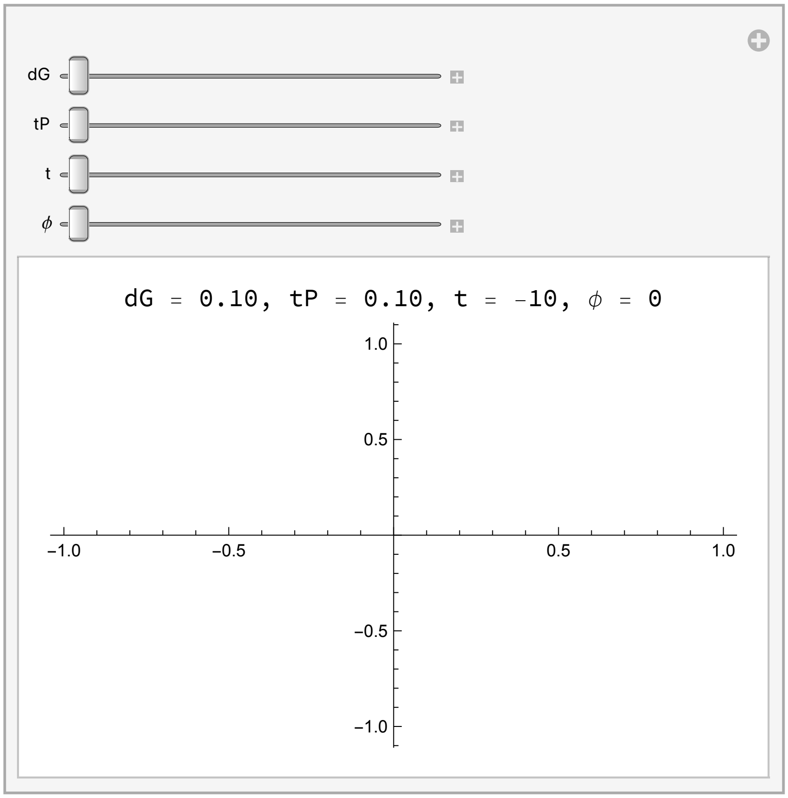
Out[ ]=



In[ ]:=
```
B[dP_, dG_, θ_, ϕ_] := If[θ ≤ M[dP, dG], ArcTan[(dP Sin[θ] Sin[ϕ])/(dG - dP Cos[θ])] + π,
  If[θ ≤ 2 π - M[dP, dG], ArcTan[(dP Sin[θ] Sin[ϕ])/(dG - dP Cos[θ])], ArcTan[(dP Sin[θ] Sin[ϕ])/(dG - dP Cos[θ])] - π]]
```

In[ ]:=
```
Manipulate[Labeled[Plot[Θ[dG, RP[dG, tP, t, θ], θ, ϕ], {θ, 0, 2 π}, PlotRange → All],
  Row[{"dG = ", NumberForm[dG, {3, 2}], ", tP = ", NumberForm[tP, {3, 2}], ", t = ", NumberForm[t, {3, 2}],
    ", ϕ = ", NumberForm[ϕ, {3, 2}]}], Top], {dG, 0.1, 50}, {tP, 0.1, 10}, {t, -10, -0.1}, {ϕ, 0, 2 π}]
```

Out[ ]=



Testing functions

In[ ]:=

```
Simplify[
```

$$\begin{pmatrix} \frac{(dG+(t-tP)\,Cos[\theta])^2}{\left(dG^2+(t-tP)^2+2\,dG\,(t-tP)\,Cos[\theta]\right)^{3/2}} & 0 & -\frac{(t-tP)\,(dG+(t-tP)\,Cos[\theta])\,Sin[\theta]}{\left(dG^2+(t-tP)^2+2\,dG\,(t-tP)\,Cos[\theta]\right)^{3/2}} \\ 0 & -\frac{1}{\sqrt{dG^2+(t-tP)^2+2\,dG\,(t-tP)\,Cos[\theta]}} & 0 \\ -\frac{(t-tP)\,(dG+(t-tP)\,Cos[\theta])\,Sin[\theta]}{\left(dG^2+(t-tP)^2+2\,dG\,(t-tP)\,Cos[\theta]\right)^{3/2}} & 0 & \frac{(t-tP)^2\,Sin[\theta]^2}{\left(dG^2+(t-tP)^2+2\,dG\,(t-tP)\,Cos[\theta]\right)^{3/2}} \end{pmatrix} /.$$

```
t → tP - dP]
```

Out[ ]=

$$\left\{\left\{\frac{(dG-dP\,Cos[\theta])^2}{\left(dG^2+dP^2-2\,dG\,dP\,Cos[\theta]\right)^{3/2}},\ 0,\ \frac{dP\,(dG-dP\,Cos[\theta])\,Sin[\theta]}{\left(dG^2+dP^2-2\,dG\,dP\,Cos[\theta]\right)^{3/2}}\right\},\right.$$
$$\left\{0,\ -\frac{1}{\sqrt{dG^2+dP^2-2\,dG\,dP\,Cos[\theta]}},\ 0\right\},$$
$$\left.\left\{\frac{dP\,(dG-dP\,Cos[\theta])\,Sin[\theta]}{\left(dG^2+dP^2-2\,dG\,dP\,Cos[\theta]\right)^{3/2}},\ 0,\ \frac{dP^2\,Sin[\theta]^2}{\left(dG^2+dP^2-2\,dG\,dP\,Cos[\theta]\right)^{3/2}}\right\}\right\}$$

In[ ]:=

```
MatrixForm[%52]
```

$$\begin{pmatrix} \frac{(dG-dP\,Cos[\theta])^2}{\left(dG^2+dP^2-2\,dG\,dP\,Cos[\theta]\right)^{3/2}} & 0 & \frac{dP\,(dG-dP\,Cos[\theta])\,Sin[\theta]}{\left(dG^2+dP^2-2\,dG\,dP\,Cos[\theta]\right)^{3/2}} \\ 0 & -\frac{1}{\sqrt{dG^2+dP^2-2\,dG\,dP\,Cos[\theta]}} & 0 \\ \frac{dP\,(dG-dP\,Cos[\theta])\,Sin[\theta]}{\left(dG^2+dP^2-2\,dG\,dP\,Cos[\theta]\right)^{3/2}} & 0 & \frac{dP^2\,Sin[\theta]^2}{\left(dG^2+dP^2-2\,dG\,dP\,Cos[\theta]\right)^{3/2}} \end{pmatrix}$$

In[ ]:=

```
MatrixForm[%50]
```

Out[ ]//MatrixForm=

$$\begin{pmatrix} \frac{(dG+x\,Cos[\theta])^2}{\left(dG^2+x^2+2\,dG\,x\,Cos[\theta]\right)^{3/2}} & 0 & -\frac{x\,(dG+x\,Cos[\theta])\,Sin[\theta]}{\left(dG^2+x^2+2\,dG\,x\,Cos[\theta]\right)^{3/2}} \\ 0 & -\frac{1}{\sqrt{dG^2+x^2+2\,dG\,x\,Cos[\theta]}} & 0 \\ -\frac{x\,(dG+x\,Cos[\theta])\,Sin[\theta]}{\left(dG^2+x^2+2\,dG\,x\,Cos[\theta]\right)^{3/2}} & 0 & \frac{x^2\,Sin[\theta]^2}{\left(dG^2+x^2+2\,dG\,x\,Cos[\theta]\right)^{3/2}} \end{pmatrix}$$

$$Simplify\left[\begin{pmatrix} \frac{(dG+x\,Cos[\theta])^2}{\left(dG^2+x^2+2\,dG\,x\,Cos[\theta]\right)^{3/2}} & 0 & -\frac{x\,(dG+x\,Cos[\theta])\,Sin[\theta]}{\left(dG^2+x^2+2\,dG\,x\,Cos[\theta]\right)^{3/2}} \\ 0 & -\frac{1}{\sqrt{dG^2+x^2+2\,dG\,x\,Cos[\theta]}} & 0 \\ -\frac{x\,(dG+x\,Cos[\theta])\,Sin[\theta]}{\left(dG^2+x^2+2\,dG\,x\,Cos[\theta]\right)^{3/2}} & 0 & \frac{x^2\,Sin[\theta]^2}{\left(dG^2+x^2+2\,dG\,x\,Cos[\theta]\right)^{3/2}} \end{pmatrix} /.\ t \to x + tP\right]$$

In[ ]:=
$$\text{Simplify}\left[t - \frac{tP^2 - 2\, dG\, tP\, \text{Cos}[\theta]}{2\, dG + 2\, tP - 2\, dG\, \text{Cos}[\theta]}\ /.\ t \to tP - dP\right]$$

Out[ ]=
$$-dP + tP + \frac{tP\, (tP - 2\, dG\, \text{Cos}[\theta])}{-2\, (dG + tP) + 2\, dG\, \text{Cos}[\theta]}$$

In[ ]:=
$$\text{TrigReduce}\left[-dP + tP + \frac{tP\, (tP - 2\, dG\, \text{Cos}[\theta])}{-2\, (dG + tP) + 2\, dG\, \text{Cos}[\theta]}\right]$$

Out[ ]=
$$\frac{-2\, dG\, dP + 2\, dG\, tP - 2\, dP\, tP + tP^2 + 2\, dG\, dP\, \text{Cos}[\theta]}{2\, (dG + tP - dG\, \text{Cos}[\theta])}$$

In[ ]:=
$$\text{Cancel}\left[\frac{-2\, dG\, dP + 2\, dG\, tP - 2\, dP\, tP + tP^2 + 2\, dG\, dP\, \text{Cos}[\theta]}{2\, (dG + tP - dG\, \text{Cos}[\theta])}\right]$$

Out[ ]=
$$\frac{2\, dG\, dP - 2\, dG\, tP + 2\, dP\, tP - tP^2 - 2\, dG\, dP\, \text{Cos}[\theta]}{2\, (-dG - tP + dG\, \text{Cos}[\theta])}$$

In[ ]:=
$$\text{FullSimplify}\left[\frac{2\, dG\, dP - 2\, dG\, tP + 2\, dP\, tP - tP^2 - 2\, dG\, dP\, \text{Cos}[\theta]}{2\, (-dG - tP + dG\, \text{Cos}[\theta])}\right]$$

Out[ ]=
$$-dP + \frac{tP\, (2\, dG + tP)}{2\, (dG + tP - dG\, \text{Cos}[\theta])}$$

Testing what variables are needed for each function. Most are only a function of t-tP, θ, and dG. Tmin is also a function of tP

In[ ]:=
$$\text{RP}[dG, tP, t, \theta]$$

Out[ ]=
$$\sqrt{dG^2 + (-t + tP)^2 - 2\, dG\, (-t + tP)\, \text{Cos}[\theta]}$$

In[ ]:=
$$t - \text{Tmin}[tG, tP, dG, \theta]$$

Out[ ]=
$$t - \frac{tP^2 - 2\, dG\, tP\, \text{Cos}[\theta]}{2\, dG + 2\, tP - 2\, dG\, \text{Cos}[\theta]}$$

In[•]:=
```
TrigReduce[t - (tP^2 - 2 dG tP Cos[θ])/(2 dG + 2 tP - 2 dG Cos[θ])]
```

Out[•]=

$$\frac{2\,dG\,t + 2\,t\,tP - tP^2 - 2\,dG\,t\,Cos[\theta] + 2\,dG\,tP\,Cos[\theta]}{2\,(dG + tP - dG\,Cos[\theta])}$$

In[•]:=
```
Simplify[t - (tP^2 - 2 dG tP Cos[θ])/(2 dG + 2 tP - 2 dG Cos[θ]) /. t → x + tP]
```

Out[•]=

$$tP + x + \frac{tP\,(tP - 2\,dG\,Cos[\theta])}{-2\,(dG + tP) + 2\,dG\,Cos[\theta]}$$

In[•]:=
```
TrigReduce[tP + x + (tP (tP - 2 dG Cos[θ]))/(-2 (dG + tP) + 2 dG Cos[θ])]
```

Out[•]=

$$\frac{2\,dG\,tP + tP^2 + 2\,dG\,x + 2\,tP\,x - 2\,dG\,x\,Cos[\theta]}{2\,(dG + tP - dG\,Cos[\theta])}$$

In[•]:=
```
Simplify[(2 dG tP + tP^2 + 2 dG x + 2 tP x - 2 dG x Cos[θ])/(2 (dG + tP - dG Cos[θ]))]
```

Out[•]=

$$\frac{2\,dG\,(tP + x) + tP\,(tP + 2\,x) - 2\,dG\,x\,Cos[\theta]}{2\,(dG + tP - dG\,Cos[\theta])}$$

In[•]:=
```
Simplify[t - (tP^2 - 2 dG tP Cos[θ])/(2 dG + 2 tP - 2 dG Cos[θ]), x = t - tP]
```

Out[•]=

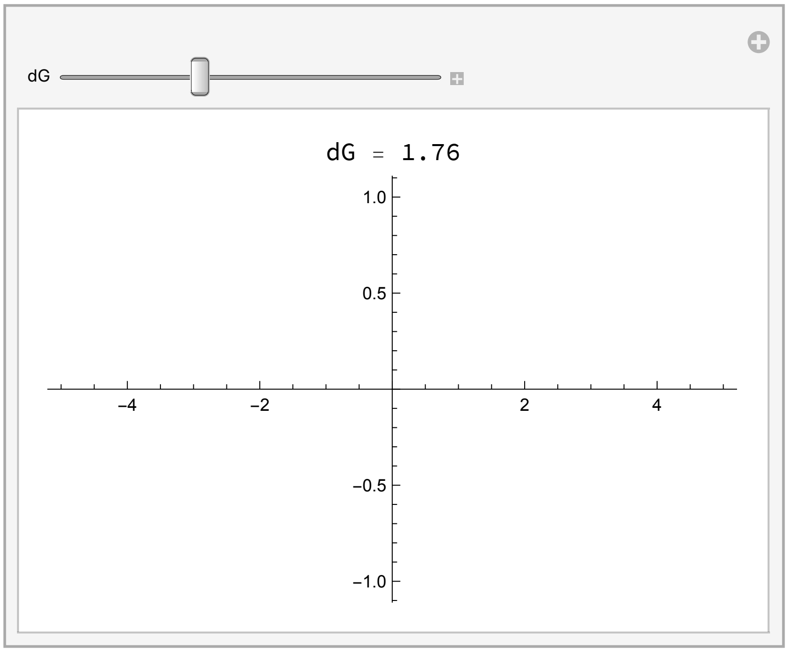
$$t + \frac{tP\,(tP - 2\,dG\,Cos[\theta])}{-2\,(dG + tP) + 2\,dG\,Cos[\theta]}$$

Graphing

In[•]:=
```
RG[dG_, t_] := If[t > (-dG), dG + t, 0];
```

In[ ]:=
```
Manipulate[Labeled[Plot[RG[dG, t], {t, -5, 5}],
  Row[{"dG = ", NumberForm[dG, {3, 2}]}], Top], {dG, 0, 5}]
```

Out[ ]=



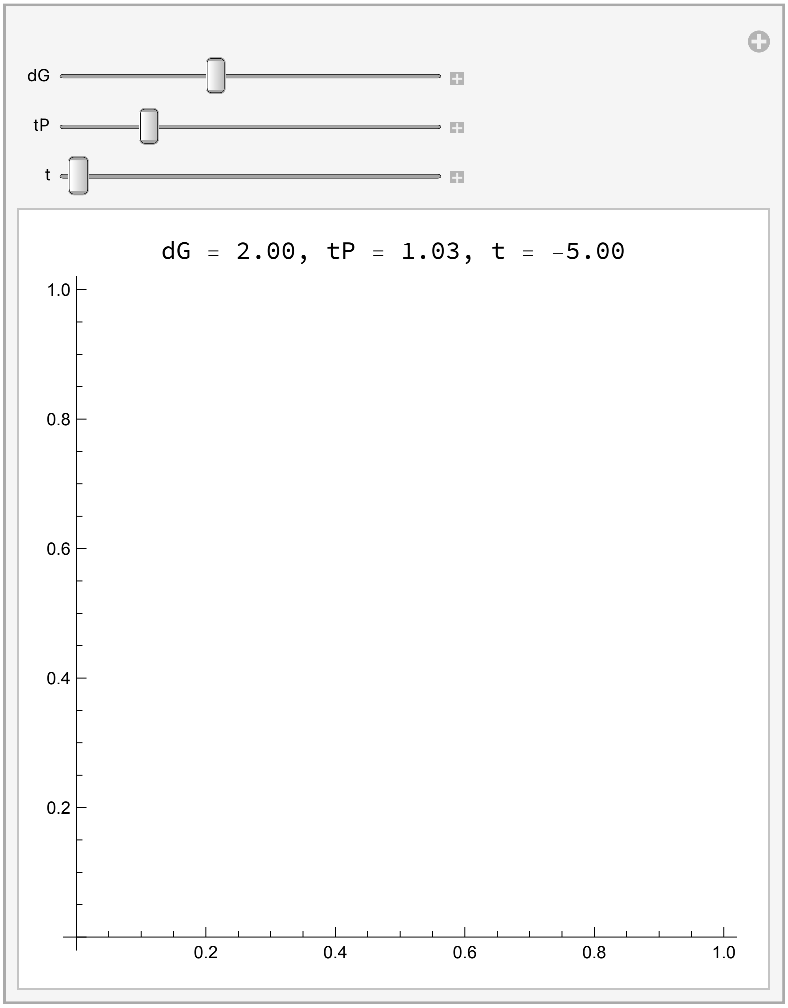
In[ ]:=
```
RP[dG, tP, t, θ]
```

Out[ ]=

$$\sqrt{dG^2 + (-t + tP)^2 - 2\, dG\, (-t + tP)\, \mathrm{Cos}[\theta]}$$

In[ ]:=
```
Manipulate[Labeled[PolarPlot[RP[dG, tP, t, θ], {θ, 0, 2 π}],
  Row[{"dG = ", NumberForm[dG, {3, 2}], ", tP = ", NumberForm[tP, {3, 2}],
   ", t = ", NumberForm[t, {3, 2}]}], Top], {dG, 0, 5}, {tP, 0, 5}, {t, -5, 5}]
```

Out[ ]=



RP works for tP = 0, dG = 1, t<0

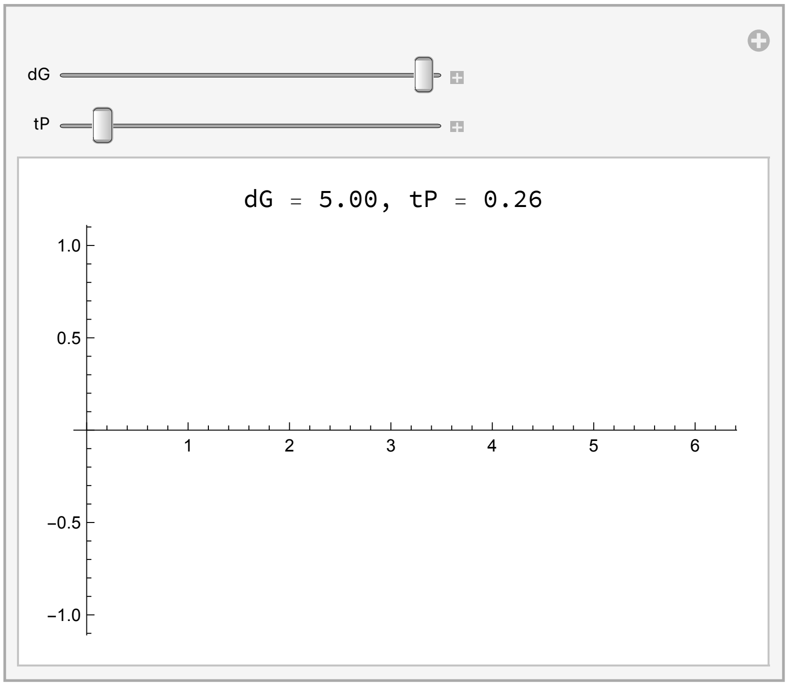
In[ ]:=
```
Tmin[tG, tP, dG, θ]
```

Out[ ]=

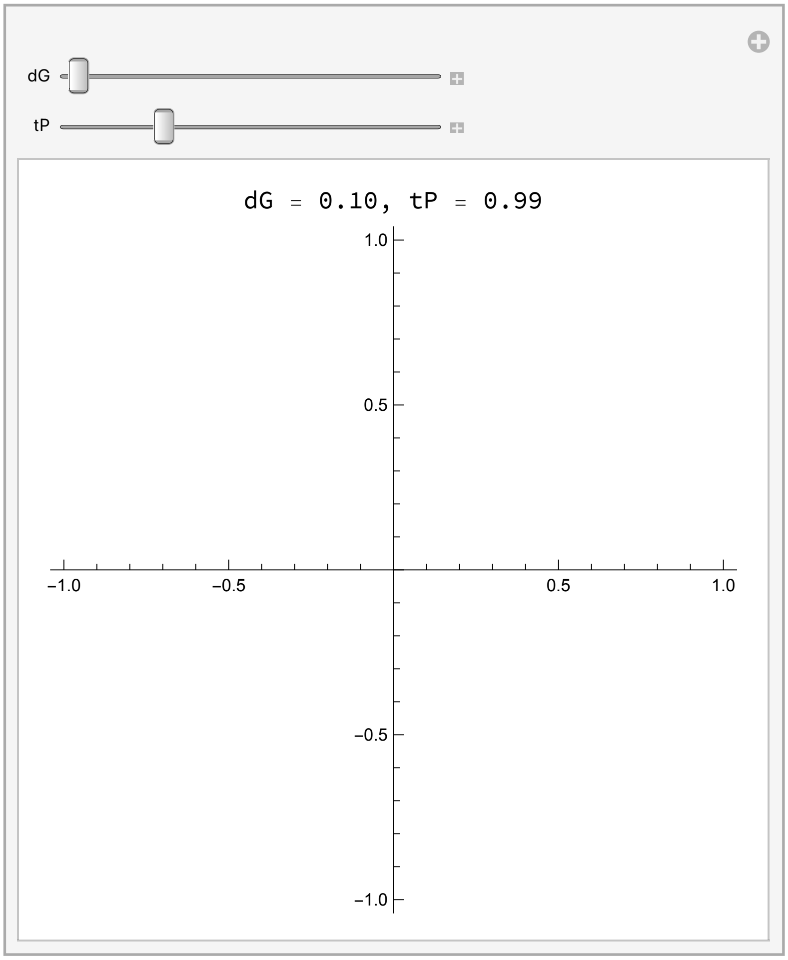
$$\frac{tP^2 - 2\,dG\,tP\,\mathrm{Cos}[\theta]}{2\,dG + 2\,tP - 2\,dG\,\mathrm{Cos}[\theta]}$$

In[∘]:=
```
Manipulate[Labeled[Plot[Tmin[tG, tP, dG, θ], {θ, 0, 2 π}],
  Row[{"dG = ", NumberForm[dG, {3, 2}], ", tP = ", NumberForm[tP, {3, 2}]}],
  Top], {dG, 0.1, 5}, {tP, 0.1, 2.5}]
```

Out[∘]=



In[ ]:=
```
Manipulate[Labeled[PolarPlot[tP - Tmin[tG, tP, dG, θ] , {θ, 0, 2 π}, PlotRange → All],
  Row[{"dG = ", NumberForm[dG, {3, 2}], ", tP = ", NumberForm[tP, {3, 2}]}],
  Top], {dG, 0.1, 30}, {tP, 0.01, 4}]
```
Out[ ]=



Ok I think this is right

+ Polarized Metric in 2D simplification

Simplifying to 2D, in the plane of the Earth, GW source, and photon. Here, ϕ=0 and only considering α.

In[ ]:=
```
P // MatrixForm
```
Out[ ]//MatrixForm=

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

In[ ]:=
```
E[α_] := Ry[-α].P.Transpose[Ry[-α]] (*Polarization of a + GW at an angle α*)
E[α] // MatrixForm
```

Out[ ]//MatrixForm=

$$\begin{pmatrix} \mathrm{Cos}[\alpha]^2 & 0 & \mathrm{Cos}[\alpha]\,\mathrm{Sin}[\alpha] \\ 0 & -1 & 0 \\ \mathrm{Cos}[\alpha]\,\mathrm{Sin}[\alpha] & 0 & \mathrm{Sin}[\alpha]^2 \end{pmatrix}$$

In[ ]:=
```
ϵ = E[A[DP[tP, t], dG, θ]]; (*plugging in α*)
FullSimplify[ϵ] // MatrixForm
```

Out[ ]//MatrixForm=

$$\begin{pmatrix} \frac{(\mathrm{dG}+(\mathrm{t}-\mathrm{tP})\,\mathrm{Cos}[\theta])^2}{\mathrm{dG}^2+(\mathrm{t}-\mathrm{tP})^2+2\,\mathrm{dG}\,(\mathrm{t}-\mathrm{tP})\,\mathrm{Cos}[\theta]} & 0 & \frac{1}{2}\,\mathrm{Sin}\left[2\,\mathrm{ArcTan}\left[\frac{(-\mathrm{t}+\mathrm{tP})\,\mathrm{Sin}[\theta]}{\mathrm{dG}+(\mathrm{t}-\mathrm{tP})\,\mathrm{Cos}[\theta]}\right]\right] \\ 0 & -1 & 0 \\ \frac{1}{2}\,\mathrm{Sin}\left[2\,\mathrm{ArcTan}\left[\frac{(-\mathrm{t}+\mathrm{tP})\,\mathrm{Sin}[\theta]}{\mathrm{dG}+(\mathrm{t}-\mathrm{tP})\,\mathrm{Cos}[\theta]}\right]\right] & 0 & \frac{(\mathrm{t}-\mathrm{tP})^2\,\mathrm{Sin}[\theta]^2}{\mathrm{dG}^2+(\mathrm{t}-\mathrm{tP})^2+2\,\mathrm{dG}\,(\mathrm{t}-\mathrm{tP})\,\mathrm{Cos}[\theta]} \end{pmatrix}$$

$$\mathrm{H}[\mathrm{t\_},\ \mathrm{tP\_},\ \mathrm{dG\_},\ \theta\_] := \frac{1}{\mathrm{RP}[\mathrm{dG},\ \mathrm{tP},\ \mathrm{t},\ \theta]}\ \mathrm{E}[\mathrm{A}[\mathrm{DP}[\mathrm{tP},\ \mathrm{t}],\ \mathrm{dG},\ \theta]]\ \text{(*not using h0*)}$$

In[ ]:= h = H[t, tP, dG, θ]; (*full expression for h_{ij}, without multiplying by h0*)

In[ ]:=
```
FullSimplify[h] // MatrixForm
```

Out[ ]//MatrixForm=

$$\begin{pmatrix} \frac{(\mathrm{dG}+(\mathrm{t}-\mathrm{tP})\,\mathrm{Cos}[\theta])^2}{(\mathrm{dG}^2+(\mathrm{t}-\mathrm{tP})^2+2\,\mathrm{dG}\,(\mathrm{t}-\mathrm{tP})\,\mathrm{Cos}[\theta])^{3/2}} & 0 & \frac{\mathrm{Sin}\left[2\,\mathrm{ArcTan}\left[\frac{(-\mathrm{t}+\mathrm{tP})\,\mathrm{Sin}[\theta]}{\mathrm{dG}+(\mathrm{t}-\mathrm{tP})\,\mathrm{Cos}[\theta]}\right]\right]}{2\,\sqrt{\mathrm{dG}^2+(\mathrm{t}-\mathrm{tP})^2+2\,\mathrm{dG}\,(\mathrm{t}-\mathrm{tP})\,\mathrm{Cos}[\theta]}} \\ 0 & -\frac{1}{\sqrt{\mathrm{dG}^2+(\mathrm{t}-\mathrm{tP})^2+2\,\mathrm{dG}\,(\mathrm{t}-\mathrm{tP})\,\mathrm{Cos}[\theta]}} & 0 \\ \frac{\mathrm{Sin}\left[2\,\mathrm{ArcTan}\left[\frac{(-\mathrm{t}+\mathrm{tP})\,\mathrm{Sin}[\theta]}{\mathrm{dG}+(\mathrm{t}-\mathrm{tP})\,\mathrm{Cos}[\theta]}\right]\right]}{2\,\sqrt{\mathrm{dG}^2+(\mathrm{t}-\mathrm{tP})^2+2\,\mathrm{dG}\,(\mathrm{t}-\mathrm{tP})\,\mathrm{Cos}[\theta]}} & 0 & \frac{(\mathrm{t}-\mathrm{tP})^2\,\mathrm{Sin}[\theta]^2}{(\mathrm{dG}^2+(\mathrm{t}-\mathrm{tP})^2+2\,\mathrm{dG}\,(\mathrm{t}-\mathrm{tP})\,\mathrm{Cos}[\theta])^{3/2}} \end{pmatrix}$$

Simplifying metric representation

Simplifying $\mathrm{Sin}\left[2\,\mathrm{ArcTan}\left[\frac{(-t+\mathrm{tP})\,\mathrm{Sin}[\theta]}{\mathrm{dG}+(t-\mathrm{tP})\,\mathrm{Cos}[\theta]}\right]\right]$

In[ ]:=
```
o = (-t + tP) Sin[θ];
a = dG + (t - tP) Cos[θ];
```

In[ ]:=
```
hyp = Sqrt[a^2 + o^2]
```

Out[ ]=

$$\sqrt{(\mathrm{dG}+(\mathrm{t}-\mathrm{tP})\,\mathrm{Cos}[\theta])^2+(-\mathrm{t}+\mathrm{tP})^2\,\mathrm{Sin}[\theta]^2}$$

In[ ]:=

$$2 * \frac{\mathrm{o}}{\mathrm{hyp}} * \frac{\mathrm{a}}{\mathrm{hyp}}\ \text{(*Sin(2x) = 2CosxSinx*)}$$

Out[ ]=

$$\frac{2\,(-\mathrm{t}+\mathrm{tP})\,(\mathrm{dG}+(\mathrm{t}-\mathrm{tP})\,\mathrm{Cos}[\theta])\,\mathrm{Sin}[\theta]}{(\mathrm{dG}+(\mathrm{t}-\mathrm{tP})\,\mathrm{Cos}[\theta])^2+(-\mathrm{t}+\mathrm{tP})^2\,\mathrm{Sin}[\theta]^2}$$

In[∘]:=
```
sin2arctan = FullSimplify[(2 (-t + tP) (dG + (t - tP) Cos[θ]) Sin[θ])/((dG + (t - tP) Cos[θ])^2 + (-t + tP)^2 Sin[θ]^2)]
 (*I graphed this against the original Sin[2ArcTan[]] and they look identical*)
```

Out[∘]=

$$-\frac{2\,(t-tP)\,(dG+(t-tP)\,\mathrm{Cos}[\theta])\,\mathrm{Sin}[\theta]}{dG^2+(t-tP)^2+2\,dG\,(t-tP)\,\mathrm{Cos}[\theta]}$$

```
cos2arctan = FullSimplify[(a/hyp)^2 - (o/hyp)^2] (*cos(2x) = cos^2(x) - sin^2(x)*)
```

Out[∘]=

$$\frac{dG^2+2\,dG\,(t-tP)\,\mathrm{Cos}[\theta]+(t-tP)^2\,\mathrm{Cos}[2\,\theta]}{dG^2+(t-tP)^2+2\,dG\,(t-tP)\,\mathrm{Cos}[\theta]}$$

```
Manipulate[Plot[-(2 (t - tP) (dG + (t - tP) Cos[θ]) Sin[θ])/(dG^2 + (t - tP)^2 + 2 dG (t - tP) Cos[θ]) - Sin[2 ArcTan[((-t + tP) Sin[θ])/(dG + (t - tP) Cos[θ])]], {t, -0.431556, 1.85832}],
 {dG, 0.1, 2.04717}, {tP, 0.1, 2}, {θ, 0, π}](*test plot, they are the same*)
```

In[∘]:=
```
(-(2 (t - tP) (dG + (t - tP) Cos[θ]) Sin[θ])/(dG^2 + (t - tP)^2 + 2 dG (t - tP) Cos[θ])) / (2 Sqrt[dG^2 + (t - tP)^2 + 2 dG (t - tP) Cos[θ]])
 (*Simplifying the expression that appears in the matrix*)
```

Out[∘]=

$$-\frac{(t-tP)\,(dG+(t-tP)\,\mathrm{Cos}[\theta])\,\mathrm{Sin}[\theta]}{\left(dG^2+(t-tP)^2+2\,dG\,(t-tP)\,\mathrm{Cos}[\theta]\right)^{3/2}}$$

In[∘]:=

$$\epsilon = \begin{pmatrix} \frac{(dG+(t-tP)\,\mathrm{Cos}[\theta])^2}{dG^2+(t-tP)^2+2\,dG\,(t-tP)\,\mathrm{Cos}[\theta]} & 0 & -\frac{(t-tP)\,(dG+(t-tP)\,\mathrm{Cos}[\theta])\,\mathrm{Sin}[\theta]}{dG^2+(t-tP)^2+2\,dG\,(t-tP)\,\mathrm{Cos}[\theta]} \\ 0 & -1 & 0 \\ -\frac{(t-tP)\,(dG+(t-tP)\,\mathrm{Cos}[\theta])\,\mathrm{Sin}[\theta]}{dG^2+(t-tP)^2+2\,dG\,(t-tP)\,\mathrm{Cos}[\theta]} & 0 & \frac{(t-tP)^2\,\mathrm{Sin}[\theta]^2}{dG^2+(t-tP)^2+2\,dG\,(t-tP)\,\mathrm{Cos}[\theta]} \end{pmatrix};$$

In[∘]:= h =

$$\begin{pmatrix} \frac{(dG+(t-tP)\,\mathrm{Cos}[\theta])^2}{\left(dG^2+(t-tP)^2+2\,dG\,(t-tP)\,\mathrm{Cos}[\theta]\right)^{3/2}} & 0 & -\frac{(t-tP)\,(dG+(t-tP)\,\mathrm{Cos}[\theta])\,\mathrm{Sin}[\theta]}{\left(dG^2+(t-tP)^2+2\,dG\,(t-tP)\,\mathrm{Cos}[\theta]\right)^{3/2}} \\ 0 & -\frac{1}{\sqrt{dG^2+(t-tP)^2+2\,dG\,(t-tP)\,\mathrm{Cos}[\theta]}} & 0 \\ -\frac{(t-tP)\,(dG+(t-tP)\,\mathrm{Cos}[\theta])\,\mathrm{Sin}[\theta]}{\left(dG^2+(t-tP)^2+2\,dG\,(t-tP)\,\mathrm{Cos}[\theta]\right)^{3/2}} & 0 & \frac{(t-tP)^2\,\mathrm{Sin}[\theta]^2}{\left(dG^2+(t-tP)^2+2\,dG\,(t-tP)\,\mathrm{Cos}[\theta]\right)^{3/2}} \end{pmatrix};$$

(*not multiplied by h0*)

In[∘]:=
```
Clear[h]
```

In[∘]:=
```
h[t_] :=
```

$$\begin{pmatrix} \frac{(dG+(t-tP)\ Cos[\theta])^2}{\left(dG^2+(t-tP)^2+2\ dG\ (t-tP)\ Cos[\theta]\right)^{3/2}} & 0 & -\frac{(t-tP)\ (dG+(t-tP)\ Cos[\theta])\ Sin[\theta]}{\left(dG^2+(t-tP)^2+2\ dG\ (t-tP)\ Cos[\theta]\right)^{3/2}} \\ 0 & -\frac{1}{\sqrt{dG^2+(t-tP)^2+2\ dG\ (t-tP)\ Cos[\theta]}} & 0 \\ -\frac{(t-tP)\ (dG+(t-tP)\ Cos[\theta])\ Sin[\theta]}{\left(dG^2+(t-tP)^2+2\ dG\ (t-tP)\ Cos[\theta]\right)^{3/2}} & 0 & \frac{(t-tP)^2\ Sin[\theta]^2}{\left(dG^2+(t-tP)^2+2\ dG\ (t-tP)\ Cos[\theta]\right)^{3/2}} \end{pmatrix}$$

In[∘]:=
```
test = h[Tmin[tG, tP, dG, θ]];
```

In[∘]:=
```
MatrixForm[test]
```

Out[∘]//MatrixForm=

$$\begin{pmatrix} \frac{\left(dG+Cos[\theta]\ \left(-tP+\frac{tP^2-2\ dG\ tP\ Cos[\theta]}{2\ dG+2\ tP-2\ dG\ Cos[\theta]}\right)\right)^2}{\left(dG^2+2\ dG\ Cos[\theta]\ \left(-tP+\frac{tP^2-2\ dG\ tP\ Cos[\theta]}{2\ dG+2\ tP-2\ dG\ Cos[\theta]}\right)+\left(-tP+\frac{tP^2-2\ dG\ tP\ Cos[\theta]}{2\ dG+2\ tP-2\ dG\ Cos[\theta]}\right)^2\right)^{3/2}} & 0 \\ 0 & -\frac{1}{\sqrt{dG^2+2\ dG\ Cos[\theta]\ \left(-tP+\frac{tP^2-2\ dG\ tP\ Cos[\theta]}{2\ dG+2\ tP-2\ dG\ Cos[\theta]}\right)+\left(-tP+\frac{tP^2-2\ dG}{2\ dG+2\ tP-}\right.}} \\ -\frac{\left(-tP+\frac{tP^2-2\ dG\ tP\ Cos[\theta]}{2\ dG+2\ tP-2\ dG\ Cos[\theta]}\right)\ \left(dG+Cos[\theta]\ \left(-tP+\frac{tP^2-2\ dG\ tP\ Cos[\theta]}{2\ dG+2\ tP-2\ dG\ Cos[\theta]}\right)\right)\ Sin[\theta]}{\left(dG^2+2\ dG\ Cos[\theta]\ \left(-tP+\frac{tP^2-2\ dG\ tP\ Cos[\theta]}{2\ dG+2\ tP-2\ dG\ Cos[\theta]}\right)+\left(-tP+\frac{tP^2-2\ dG\ tP\ Cos[\theta]}{2\ dG+2\ tP-2\ dG\ Cos[\theta]}\right)^2\right)^{3/2}} & 0 \end{pmatrix}$$

```
ξidk = FullSimplify[n[θ].Transpose[n[θ].test]] (*units of 1/length?*)
```

Out[∘]=

$$\frac{8\ dG^2\ (dG+tP-dG\ Cos[\theta])^3\ Sin[\theta]^2}{\left(2\ dG^2+2\ dG\ tP+tP^2-2\ dG\ (dG+tP)\ Cos[\theta]\right)^3}$$

In[∘]:=
```
RTmin = FullSimplify[RP[dG, tP, Tmin[tG, tP, dG, θ], θ]]
```

Out[∘]=

$$dG+tP-\frac{tP\ (2\ dG+tP)}{2\ (dG+tP-dG\ Cos[\theta])}$$

$$\mathrm{SphericalPlot3D}\left[\frac{8\ dG^2\ (dG+tP-dG\ Cos[\theta])^3\ Sin[\theta]^2}{\left(2\ dG^2+2\ dG\ tP+tP^2-2\ dG\ (dG+tP)\ Cos[\theta]\right)^3},\ \{\theta,\ 0,\ 2\ \pi\}\right],$$

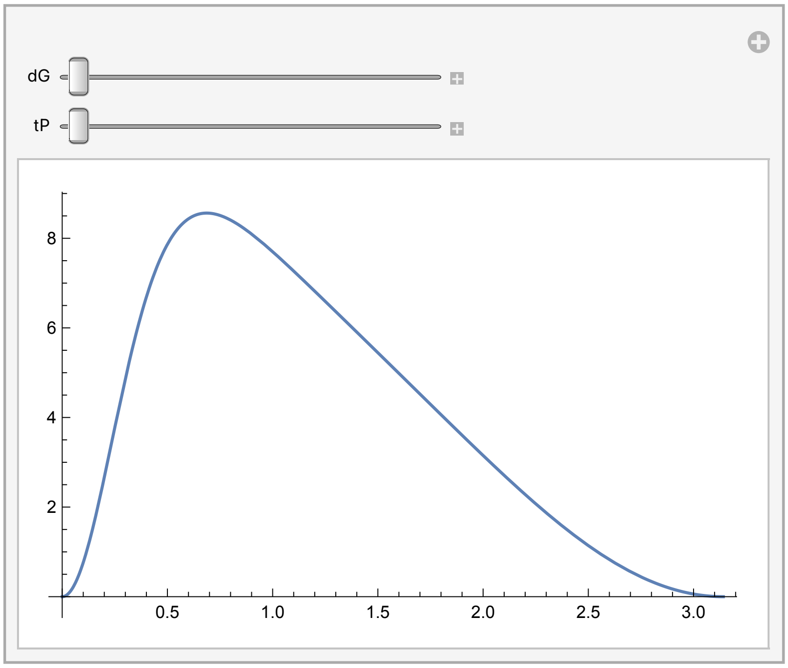
```
 {dG, -1.237598476615519`, 1.237598476615519`},
 {tP, -1.4072596541852507`, 1.4072596541852507`}]
```

In[ ]:=

```
Manipulate[Plot[(8 dG^2 (dG + tP - dG Cos[θ])^3 Sin[θ]^2)/(2 dG^2 + 2 dG tP + tP^2 - 2 dG (dG + tP) Cos[θ])^3,
  {θ, 0, π}, PlotRange → All], {dG, 0.1, 5}, {tP, 0.1, 5}]
```

$$\text{Manipulate}\left[\text{Plot}\left[\frac{8\,dG^2\,(dG + tP - dG\,\text{Cos}[\theta])^3\,\text{Sin}[\theta]^2}{\left(2\,dG^2 + 2\,dG\,tP + tP^2 - 2\,dG\,(dG + tP)\,\text{Cos}[\theta]\right)^3},\ \{\theta, 0, \pi\},\ \text{PlotRange} \to \text{All}\right],\ \{dG, 0.1, 5\},\ \{tP, 0.1, 5\}\right]$$

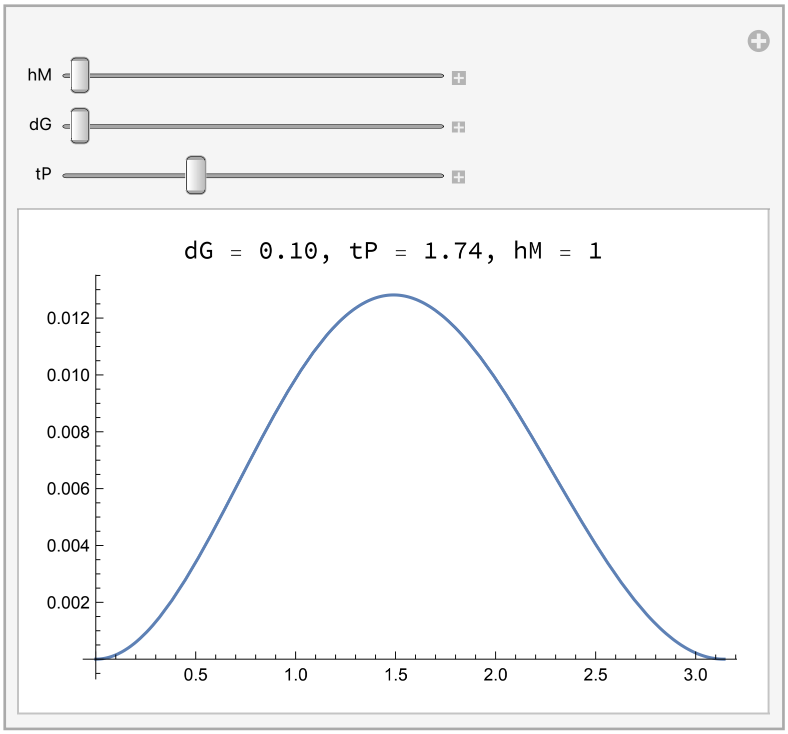
Out[ ]=



In[∘]:=

```
Manipulate[
 Labeled[Plot[(8 dG^2 (dG + tP - dG Cos[θ])^3 Sin[θ]^2)/(2 dG^2 + 2 dG tP + tP^2 - 2 dG (dG + tP) Cos[θ])^3, {θ, 0, π}, PlotRange → All],
  Row[{"dG = ", NumberForm[dG, {3, 2}], ", tP = ", NumberForm[tP, {3, 2}], ", hM = ",
    NumberForm[hM, {3, 2}]}], Top], {hM, 1, 2}, {dG, 0.1, 5}, {tP, 0.1, 5}]
```

Out[∘]=



In[∘]:=

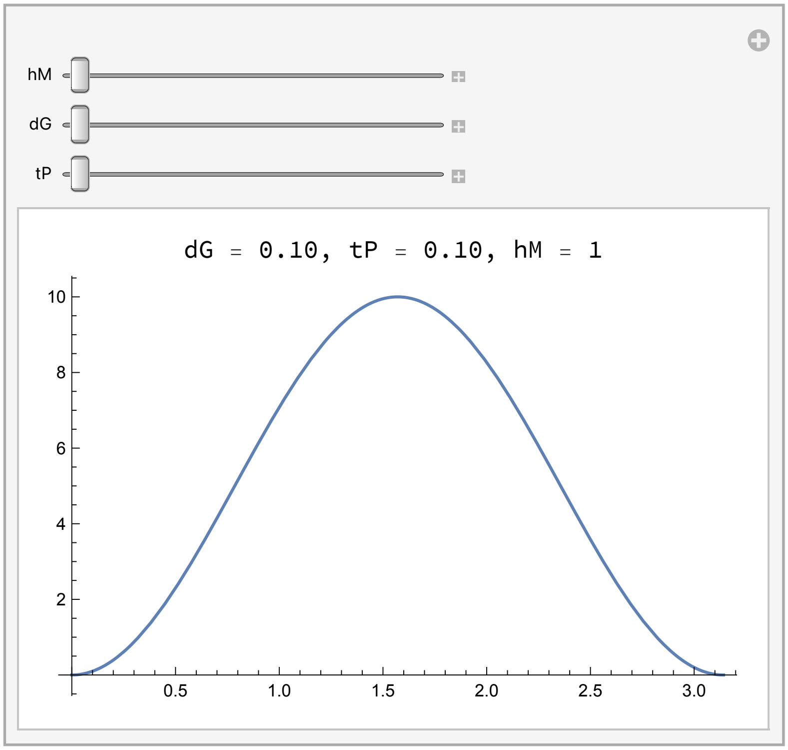
$$\frac{dG^2\,\mathrm{Sin}[\theta]^2}{\left(dG^2 + (tP - tP)^2 + 2\,dG\,(tP - tP)\,\mathrm{Cos}[\theta]\right)^{3/2}}$$

Out[∘]=

$$\frac{\mathrm{Sin}[\theta]^2}{\sqrt{dG^2}}$$

In[∘]:= Manipulate[Labeled[Plot[$\dfrac{\text{Sin}[\theta]^2}{\sqrt{\text{dG}^2}}$, {θ, 0, π}, PlotRange → All],
Row[{"dG = ", NumberForm[dG, {3, 2}], ", tP = ", NumberForm[tP, {3, 2}], ", hM = ",
NumberForm[hM, {3, 2}]}], Top], {hM, 1, 2}, {dG, 0.1, 5}, {tP, 0.1, 5}]

Out[∘]=



In[∘]:= MatrixForm[h[tP]]

Out[∘]//MatrixForm=

$$\begin{pmatrix} \frac{1}{\sqrt{\text{dG}^2}} & 0 & 0 \\ 0 & -\frac{1}{\sqrt{\text{dG}^2}} & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

In[∘]:= FullSimplify[n[θ].Transpose[n[θ].h[tP]]]

Out[∘]=

$$\frac{\text{Sin}[\theta]^2}{\text{dG}}$$

+ Polarized Redshift in 2D

GW source is orbiting in the y/z plane. We're looking at a photon that has been received in the x/z plane (perpendicular to the source's orbital plane.

In[∘]:=
```
n[θ_] := {Sin[θ], 0, Cos[θ]} (*2D version in x/z plane*)
FullSimplify[n[θ].Transpose[n[θ].h]] (*applying n^i n^j*) (*not multiplied by h0*)
```

Out[∘]=

$$\frac{dG^2 \, Sin[\theta]^2}{\left(dG^2 + (t - tP)^2 + 2\, dG\, (t - tP)\, Cos[\theta]\right)^{3/2}}$$

In[∘]:=

$$\frac{dG^2\, Sin[\theta]^2}{\left(dG^2 + (Tmin[tG, tP, dG, \theta] - tP)^2 + 2\, dG\, (Tmin[tG, tP, dG, \theta] - tP)\, Cos[\theta]\right)^{3/2}} - \frac{dG^2\, Sin[\theta]^2}{\left(dG^2 + (tP - tP)^2 + 2\, dG\, (tP - tP)\, Cos[\theta]\right)^{3/2}}$$

```
(*taking the difference manually, updated tmin*)
```

Out[∘]=

$$-\frac{Sin[\theta]^2}{\sqrt{dG^2}} + \frac{dG^2\, Sin[\theta]^2}{\left(dG^2 + 2\, dG\, Cos[\theta]\left(-tP + \frac{tP^2 - 2\, dG\, tP\, Cos[\theta]}{2\, dG + 2\, tP - 2\, dG\, Cos[\theta]}\right) + \left(-tP + \frac{tP^2 - 2\, dG\, tP\, Cos[\theta]}{2\, dG + 2\, tP - 2\, dG\, Cos[\theta]}\right)^2\right)^{3/2}}$$

In[∘]:=

$$FullSimplify\left[-\frac{Sin[\theta]^2}{\sqrt{dG^2}} + \frac{dG^2\, Sin[\theta]^2}{\left(dG^2 + 2\, dG\, Cos[\theta]\left(-tP + \frac{tP^2 - 2\, dG\, tP\, Cos[\theta]}{2\, dG + 2\, tP - 2\, dG\, Cos[\theta]}\right) + \left(-tP + \frac{tP^2 - 2\, dG\, tP\, Cos[\theta]}{2\, dG + 2\, tP - 2\, dG\, Cos[\theta]}\right)^2\right)^{3/2}}\right]$$

Out[∘]=

$$\left(-\frac{1}{dG} + \frac{8\, dG^2\, (dG + tP - dG\, Cos[\theta])^3}{\left(2\, dG^2 + 2\, dG\, tP + tP^2 - 2\, dG\, (dG + tP)\, Cos[\theta]\right)^3}\right) Sin[\theta]^2$$

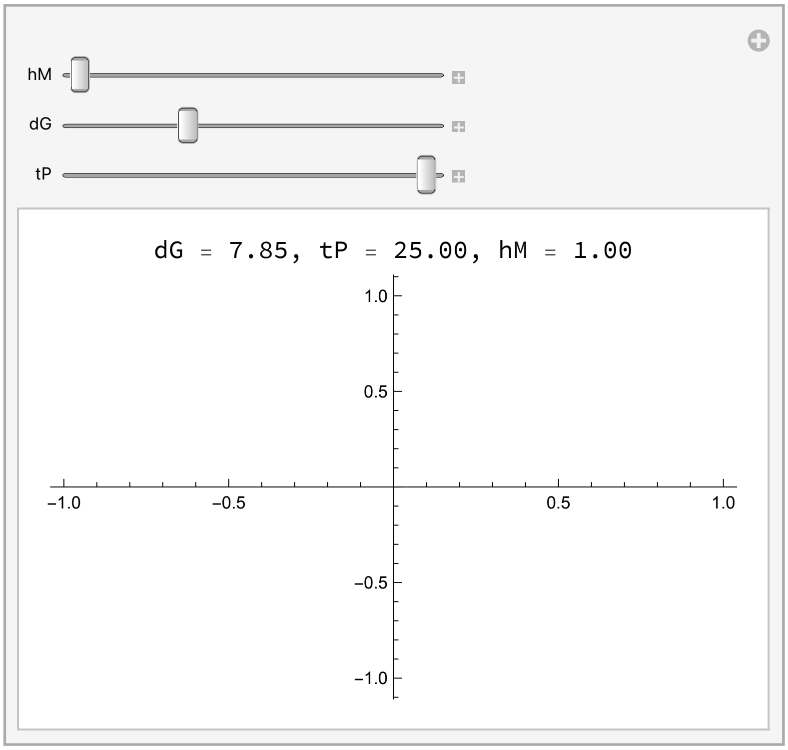
z is this difference times -0.5*h0 (since we have not yet multiplied by h0) and h0 = hM*dG. The units are a big weird. hM is unit-less because it's the measured strain, but h0 is the hypothetical original strain, which has units of distance from its scaling factor. In total, redshift is unit-less as we'd expect.

$$Z[\theta\_, dG\_, tP\_, hM\_] := -\frac{hM * dG}{2}\left(-\frac{1}{dG} + \frac{8\, dG^2\, (dG + tP - dG\, Cos[\theta])^3}{\left(2\, dG^2 + 2\, dG\, tP + tP^2 - 2\, dG\, (dG + tP)\, Cos[\theta]\right)^3}\right) Sin[\theta]^2$$

```
(*find denominator roots*)
```

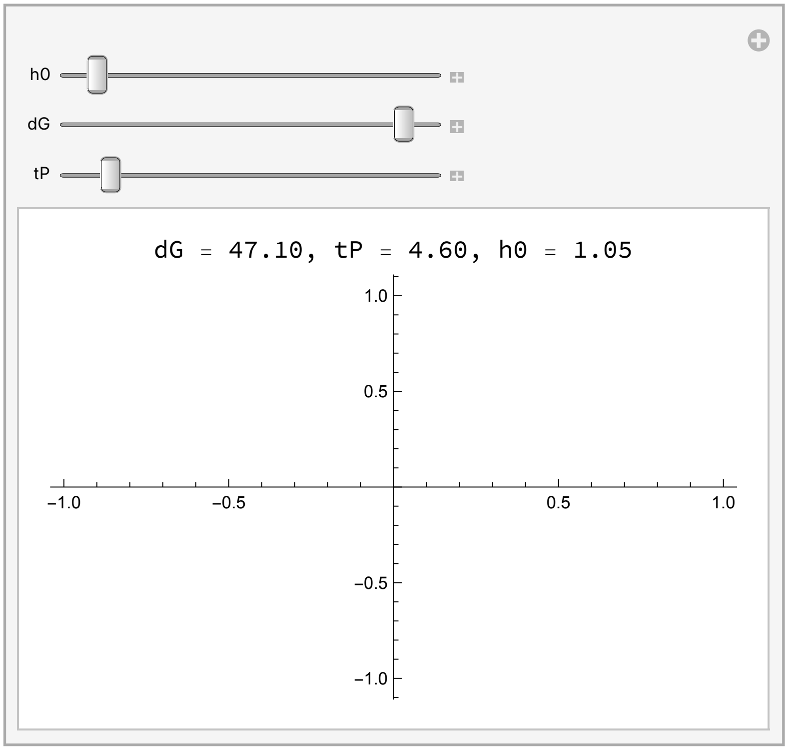
In[ ]:=
```
Manipulate[Labeled[Plot[Z[θ * π, dG, tP, hM], {θ, 0, 1}, PlotRange → All],
  Row[{"dG = ", NumberForm[dG, {3, 2}], ", tP = ", NumberForm[tP, {3, 2}], ", hM = ",
    NumberForm[hM, {3, 2}]}], Top], {hM, 1, 2}, {dG, 0.1, 25}, {tP, 0.01, 25}]
```

Out[ ]=



In[∘]:=
```
Manipulate[Labeled[Plot[Z[θ, dG, tP, h0/dG], {θ, 0, π}, PlotRange → All],
  Row[{"dG = ", NumberForm[dG, {3, 2}], ", tP = ", NumberForm[tP, {3, 2}], ", h0 = ",
    NumberForm[h0, {3, 2}]}], Top], {h0, 1, 2}, {dG, 0.1, 50}, {tP, 0.01, 50}]
```

Out[∘]=



+ Polarized Deflection in 2D

The following code is copied from the formal thesis writeup Mathematica document

```
n[θ_] := {Sin[θ], 0, Cos[θ]} (*2D looking vectore in x/z plane*)
```

In[∘]:=
```
HSimp[t_] :=
```

$$\begin{pmatrix} \frac{(dG+(t-tP)\,Cos[\theta])^2}{(dG^2+(t-tP)^2+2\,dG\,(t-tP)\,Cos[\theta])^{3/2}} & 0 & -\frac{(t-tP)\,(dG+(t-tP)\,Cos[\theta])\,Sin[\theta]}{(dG^2+(t-tP)^2+2\,dG\,(t-tP)\,Cos[\theta])^{3/2}} \\ 0 & -\frac{1}{\sqrt{dG^2+(t-tP)^2+2\,dG\,(t-tP)\,Cos[\theta]}} & 0 \\ -\frac{(t-tP)\,(dG+(t-tP)\,Cos[\theta])\,Sin[\theta]}{(dG^2+(t-tP)^2+2\,dG\,(t-tP)\,Cos[\theta])^{3/2}} & 0 & \frac{(t-tP)^2\,Sin[\theta]^2}{(dG^2+(t-tP)^2+2\,dG\,(t-tP)\,Cos[\theta])^{3/2}} \end{pmatrix}$$

In[∘]:=
```
nnHSimp[t_] := FullSimplify[n[θ].Transpose[n[θ].HSimp[t]]] (*applying n^i n^j*)
```

In[ ]:=
```
T1 = FullSimplify[n[θ].HSimp[tP]] (*applying ninj*)
```
Out[ ]=
$$\left\{\frac{\text{Sin}[\theta]}{\text{dG}}, 0, 0\right\}$$

In[ ]:=
```
T2 = FullSimplify[n[θ] * nnHSimp[tP]]
```
Out[ ]=
$$\left\{\frac{\text{Sin}[\theta]^3}{\text{dG}}, 0, \frac{\text{Cos}[\theta]\ \text{Sin}[\theta]^2}{\text{dG}}\right\}$$

In[ ]:=
```
nnHSimp[tP] (*λ=0*)
```
Out[ ]=
$$\frac{\text{Sin}[\theta]^2}{\text{dG}}$$

(New angle approach) Plus Polarized Metric in 3D

Need to double check formula for B and A

β

In[ ]:=
```
hp3d = FullSimplify[
   Rx[B[DP[tP, t], dG, θ, ϕ]].Ry[-A[DP[tP, t], dG, θ, ϕ]].P/RP[dG, tP, t, θ].
    Transpose[Ry[-A[DP[tP, t], dG, θ, ϕ]]].Transpose[Rx[B[DP[tP, t], dG, θ, ϕ]]]];
MatrixForm[hp3d]
```

Out[ ]//MatrixForm=

$$\begin{pmatrix} \frac{(dG+(t-tP)\,Cos[\theta])^2}{\sqrt{dG^2+(t-tP)^2+2\,dG\,(t-tP)\,Cos[\theta]}\;\left((dG+(t-tP)\,Cos[\theta])^2+(t-tP)^2\,Cos[\phi]^2\,Sin[\theta]^2\right)} \\ \begin{cases} \left((-t+tP)\,Sin[\theta]\,Sin[\phi]\,Sin\left[2\,ArcTan\left[\frac{(-t+tP)\,Cos[\phi]\,Sin[\theta]}{dG+(t-tP)\,Cos[\theta]}\right]\right]\right)\Big/\left(2\,(dG+(t-tP)\,Cos[\theta])\sqrt{\left(\left(dG^2+(t-tP)^2+2\,dG\,(t-tP)\,Cos[\theta]\right)\left(1+\frac{(t-tP)^2\,Sin[\theta]^2\,Sin[\phi]^2}{(dG+(t-tP)\,Cos[\theta])^2}\right)\right)}\right) & \left(dG+t\geq tP\;||\;\theta\leq ArcCos\left[\frac{dG}{-t+tP}\right]\;||\;\theta+ArcCos\left[\frac{dG}{-t+tP}\right]>2\,\pi\right)\;\&\&\;(dG+t<tP\;||\;\theta\leq 0) \\ \left((t-tP)\,Sin[\theta]\,Sin[\phi]\,Sin\left[2\,ArcTan\left[\frac{(-t+tP)\,Cos[\phi]\,Sin[\theta]}{dG+(t-tP)\,Cos[\theta]}\right]\right]\right)\Big/\left(2\,(dG+(t-tP)\,Cos[\theta])\sqrt{\left(\left(dG^2+(t-tP)^2+2\,dG\,(t-tP)\,Cos[\theta]\right)\left(1+\frac{(t-tP)^2\,Sin[\theta]^2\,Sin[\phi]^2}{(dG+(t-tP)\,Cos[\theta])^2}\right)\right)}\right) & True \end{cases} \\ \begin{cases} Sin\left[2\,ArcTan\left[\frac{(t-tP)\,Cos[\phi]\,Sin[\theta]}{dG+(t-tP)\,Cos[\theta]}\right]\right]\Big/\left(2\sqrt{\left(\left(dG^2+(t-tP)^2+2\,dG\,(t-tP)\,Cos[\theta]\right)\left(1+\frac{(t-tP)^2\,Sin[\theta]^2\,Sin[\phi]^2}{(dG+(t-tP)\,Cos[\theta])^2}\right)\right)}\right) & \left(dG+t\geq tP\;||\;\theta\leq ArcCos\left[\frac{dG}{-t+tP}\right]\;||\;\theta+ArcCos\left[\frac{dG}{-t+tP}\right]>2\,\pi\right)\;\&\&\;(dG+t<tP\;||\;\theta\leq 0) \\ Sin\left[2\,ArcTan\left[\frac{(-t+tP)\,Cos[\phi]\,Sin[\theta]}{dG+(t-tP)\,Cos[\theta]}\right]\right]\Big/\left(2\sqrt{\left(\left(dG^2+(t-tP)^2+2\,dG\,(t-tP)\,Cos[\theta]\right)\left(1+\frac{(t-tP)^2\,Sin[\theta]^2\,Sin[\phi]^2}{(dG+(t-tP)\,Cos[\theta])^2}\right)\right)}\right) & True \end{cases} \end{pmatrix}$$

In[ ]:=
$$\text{hp3d1simp} = \left(\begin{array}{ll} \frac{(dG+(t-tP)\,\text{Cos}[\theta])^2}{\sqrt{dG^2+(t-tP)^2+2\,dG\,(t-tP)\,\text{Cos}[\theta]}\,\left((dG+(t-tP)\,\text{Cos}[\theta])^2+(t-tP)^2\,\text{Cos}[\phi]^2\,\text{Sin}[\theta]^2\right)} & \frac{(-t+tP)\,\text{Sin}}{2\,(dG+(t-tP)\,\text{Cc}} \\ \frac{(-t+tP)\,\text{Sin}[\theta]\,\text{Sin}[\phi]\,\text{TrigSimp}[(-t+tP)\,\text{Cos}[\phi]\,\text{Sin}[\theta],dG+(t-tP)\,\text{Cos}[\theta]]}{2\,(dG+(t-tP)\,\text{Cos}[\theta])\,\sqrt{\left(dG^2+(t-tP)^2+2\,dG\,(t-tP)\,\text{Cos}[\theta]\right)\,\left(1+\frac{(t-tP)^2\,\text{Sin}[\theta]^2\,\text{Sin}[\phi]^2}{(dG+(t-tP)\,\text{Cos}[\theta])^2}\right)}} & \frac{-(}{\sqrt{dG^2+(t-tP)^2+}} \\ \frac{\text{TrigSimp}[(t-tP)\,\text{Cos}[\phi]\,\text{Sin}[\theta],dG+(t-tP)\,\text{Cos}[\theta]]}{2\,\sqrt{\left(dG^2+(t-tP)^2+2\,dG\,(t-tP)\,\text{Cos}[\theta]\right)\,\left(1+\frac{(t-tP)^2\,\text{Sin}[\theta]^2\,\text{Sin}[\phi]^2}{(dG+(t-tP)\,\text{Cos}[\theta])^2}\right)}} & \frac{((dG+(t-tP)\,\text{Cos}[\theta])^2+2}{2\,\sqrt{dG^2+(t-tP)^2+}} \end{array}\right.$$

In[ ]:=
$$\text{hp3d2simp} = \left(\begin{array}{ll} \frac{(dG+(t-tP)\,\text{Cos}[\theta])^2}{\sqrt{dG^2+(t-tP)^2+2\,dG\,(t-tP)\,\text{Cos}[\theta]}\,\left((dG+(t-tP)\,\text{Cos}[\theta])^2+(t-tP)^2\,\text{Cos}[\phi]^2\,\text{Sin}[\theta]^2\right)} & \frac{(}{2\,(dG+(t-tP)\,\text{Cc}} \\ \frac{(t-tP)\,\text{Sin}[\theta]\,\text{Sin}[\phi]\,\text{TrigSimp}[(-t+tP)\,\text{Cos}[\phi]\,\text{Sin}[\theta],dG+(t-tP)\,\text{Cos}[\theta]]}{2\,(dG+(t-tP)\,\text{Cos}[\theta])\,\sqrt{\left(dG^2+(t-tP)^2+2\,dG\,(t-tP)\,\text{Cos}[\theta]\right)\,\left(1+\frac{(t-tP)^2\,\text{Sin}[\theta]^2\,\text{Sin}[\phi]^2}{(dG+(t-tP)\,\text{Cos}[\theta])^2}\right)}} & \frac{-(}{\sqrt{dG^2+(t-tP)^2+}} \\ \frac{\text{TrigSimp}[(-t+tP)\,\text{Cos}[\phi]\,\text{Sin}[\theta],dG+(t-tP)\,\text{Cos}[\theta]]}{2\,\sqrt{\left(dG^2+(t-tP)^2+2\,dG\,(t-tP)\,\text{Cos}[\theta]\right)\,\left(1+\frac{(t-tP)^2\,\text{Sin}[\theta]^2\,\text{Sin}[\phi]^2}{(dG+(t-tP)\,\text{Cos}[\theta])^2}\right)}} & \frac{((dG+(t-tP)\,\text{Cos}[\theta])^2+2}{2\,\sqrt{dG^2+(t-tP)^2+}} \end{array}\right.$$

Original

$$\left(dG + t \geq tP \;||\; \theta \leq \text{ArcCos}\left[\frac{dG}{-t+tP}\right] \;||\; \theta + \text{ArcCos}\left[\frac{dG}{-t+tP}\right] > 2\,\pi\right) \,\&\&\, (dG + t < tP \;||\; \theta \leq 0)$$

General simplification $0 < \theta < 2\pi$

$$\left(\theta \leq \text{ArcCos}\left[\frac{dG}{-t+tP}\right] \;||\; \theta + \text{ArcCos}\left[\frac{dG}{-t+tP}\right] > 2\,\pi\right) \,\&\&\, (dG + t < tP)$$

For t = tP this is False

In[ ]:= `- Zint2[tP]`

For t = Tmin

In[ ]:= `Tmin[tG, tP, dG, θ]`

Out[ ]=
$$\frac{tP^2 - 2\,dG\,tP\,\text{Cos}[\theta]}{2\,dG + 2\,tP - 2\,dG\,\text{Cos}[\theta]}$$

In[ ]:=
$$\left(\theta \leq \text{ArcCos}\left[\frac{dG}{-\text{Tmin}[tG,\ tP,\ dG,\ \theta] + tP}\right] \;||\; \theta + \text{ArcCos}\left[\frac{dG}{-\text{Tmin}[tG,\ tP,\ dG,\ \theta] + tP}\right] > 2\,\pi\right) \,\&\&\, (dG + \text{Tmin}[tG,\ tP,\ dG,\ \theta] < tP)$$

$$\left(\left(\theta \le \text{ArcCos}\left[\frac{dG}{tP - \frac{tP^2 - 2\,dG\,tP\,\text{Cos}[\theta]}{2\,dG + 2\,tP - 2\,dG\,\text{Cos}[\theta]}}\right] \,||\, \theta + \text{ArcCos}\left[\frac{dG}{tP - \frac{tP^2 - 2\,dG\,tP\,\text{Cos}[\theta]}{2\,dG + 2\,tP - 2\,dG\,\text{Cos}[\theta]}}\right] > 2\,\pi\right) \&\&\right.$$

$$\left. dG + \frac{tP^2 - 2\,dG\,tP\,\text{Cos}[\theta]}{2\,dG + 2\,tP - 2\,dG\,\text{Cos}[\theta]} < tP\right)$$

In[∘]:=
```
Zint1[Tmin[tG, tP, dG, θ]]
```

In[∘]:=

$$\left(\text{True} \,||\, \theta \le \text{ArcCos}\left[\frac{dG}{-t + tP}\right] \,||\, \theta + \text{ArcCos}\left[\frac{dG}{-t + tP}\right] > 2\,\pi\right) \&\& (\text{False} \,||\, \theta \le 0)$$

Out[∘]=

$$\theta \le 0$$

In[∘]:=

$$\begin{cases} 3 + 2 & \left(2 < 1 \,||\, \theta \le \text{ArcCos}\left[\frac{dG}{-t+tP}\right] \,||\, \theta + \text{ArcCos}\left[\frac{dG}{-t+tP}\right] > 2\,\pi\right) \&\& (3 == 4 \,||\, -1 \le 0) \\ 3 - 2 & \text{True} \end{cases}$$

Out[∘]=

$$\begin{cases} 5 & \theta \le \text{ArcCos}\left[\frac{dG}{-t+tP}\right] \,||\, \theta + \text{ArcCos}\left[\frac{dG}{-t+tP}\right] > 2\,\pi \\ 1 & \text{True} \end{cases}$$

In[∘]:=
```
v[θ_, ϕ_] := {Sin[θ] Cos[ϕ], Sin[θ] Sin[ϕ], Cos[θ]} (*3D version *)
```

First case, condition

$$\left(dG + t \ge tP \,||\, \theta \le \text{ArcCos}\left[\frac{dG}{-t+tP}\right] \,||\, \theta + \text{ArcCos}\left[\frac{dG}{-t+tP}\right] > 2\,\pi\right) \&\& (dG + t < tP \,||\, \theta \le 0$$

In[∘]:=
```
Simplify[v[θ, ϕ].Transpose[v[θ, ϕ].hp3d1simp]]
```

Out[∘]=

$$\left(\text{Cos}[\phi]\,\text{Sin}[\theta]\,\left((dG + (t - tP)\,\text{Cos}[\theta])^2 + (t - tP)^2\,\text{Sin}[\theta]^2\,\text{Sin}[\phi]^2\right)\right.$$

$$\left((t - tP)\,\text{Cos}[\theta]\,(dG + (t - tP)\,\text{Cos}[\theta])\,\sqrt{dG^2 + (t - tP)^2 + 2\,dG\,(t - tP)\,\text{Cos}[\theta]}\right.$$

$$\text{Cos}[\phi]\,\text{Sin}[\theta] + (t - tP)^2\,\sqrt{dG^2 + (t - tP)^2 + 2\,dG\,(t - tP)\,\text{Cos}[\theta]}$$

$$\text{Cos}[\phi]\,\text{Sin}[\theta]^3\,\text{Sin}[\phi]^2 + (dG + (t - tP)\,\text{Cos}[\theta])^2\,\text{Cos}[\phi]\,\text{Sin}[\theta]$$

$$\left.\sqrt{\left(dG^2 + (t - tP)^2 + 2\,dG\,(t - tP)\,\text{Cos}[\theta]\right)\,\left(1 + \frac{(t - tP)^2\,\text{Sin}[\theta]^2\,\text{Sin}[\phi]^2}{(dG + (t - tP)\,\text{Cos}[\theta])^2}\right)}\right) +$$

$$\mathrm{Sin}[\theta]\ \mathrm{Sin}[\phi]\ \Big((t-tP)^2\ \sqrt{dG^2+(t-tP)^2+2\ dG\ (t-tP)\ \mathrm{Cos}[\theta]}\ \mathrm{Cos}[\phi]^2$$

$$\mathrm{Sin}[\theta]^3\ \mathrm{Sin}[\phi]\ \left((dG+(t-tP)\ \mathrm{Cos}[\theta])^2+(t-tP)^2\ \mathrm{Sin}[\theta]^2\ \mathrm{Sin}[\phi]^2\right)+$$

$$(t-tP)\ \mathrm{Cos}[\theta]\ (dG+(t-tP)\ \mathrm{Cos}[\theta])\ \mathrm{Sin}[\theta]$$

$$\left((dG+(t-tP)\ \mathrm{Cos}[\theta])^2+2\ (t-tP)^2\ \mathrm{Cos}[\phi]^2\ \mathrm{Sin}[\theta]^2\right)\ \mathrm{Sin}[\phi]$$

$$\sqrt{\left(dG^2+(t-tP)^2+2\ dG\ (t-tP)\ \mathrm{Cos}[\theta]\right)\ \left(1+\frac{(t-tP)^2\ \mathrm{Sin}[\theta]^2\ \mathrm{Sin}[\phi]^2}{(dG+(t-tP)\ \mathrm{Cos}[\theta])^2}\right)}\ -\mathrm{Sin}[\theta]$$

$$\mathrm{Sin}[\phi]\ \sqrt{\left(dG^2+(t-tP)^2+2\ dG\ (t-tP)\ \mathrm{Cos}[\theta]\right)\ \left(1+\frac{(t-tP)^2\ \mathrm{Sin}[\theta]^2\ \mathrm{Sin}[\phi]^2}{(dG+(t-tP)\ \mathrm{Cos}[\theta])^2}\right)}$$

$$\Big((dG+(t-tP)\ \mathrm{Cos}[\theta])^4+(t-tP)^2\ (dG+(t-tP)\ \mathrm{Cos}[\theta])^2\ \mathrm{Cos}[\phi]^2\ \mathrm{Sin}[\theta]^2-$$

$$(t-tP)^4\ \mathrm{Cos}[\phi]^2\ \mathrm{Sin}[\theta]^4\ \mathrm{Sin}[\phi]^2\Big)\Big)+$$

$$\mathrm{Cos}[\theta]\ \Big((t-tP)\ (dG+(t-tP)\ \mathrm{Cos}[\theta])\ \sqrt{dG^2+(t-tP)^2+2\ dG\ (t-tP)\ \mathrm{Cos}[\theta]}$$

$$\mathrm{Cos}[\phi]^2\ \mathrm{Sin}[\theta]^2\ \left((dG+(t-tP)\ \mathrm{Cos}[\theta])^2+(t-tP)^2\ \mathrm{Sin}[\theta]^2\ \mathrm{Sin}[\phi]^2\right)+(t-tP)$$

$$(dG+(t-tP)\ \mathrm{Cos}[\theta])\ \mathrm{Sin}[\theta]^2\ \left((dG+(t-tP)\ \mathrm{Cos}[\theta])^2+2\ (t-tP)^2\ \mathrm{Cos}[\phi]^2\ \mathrm{Sin}[\theta]^2\right)$$

$$\mathrm{Sin}[\phi]^2\ \sqrt{\left(dG^2+(t-tP)^2+2\ dG\ (t-tP)\ \mathrm{Cos}[\theta]\right)\ \left(1+\frac{(t-tP)^2\ \mathrm{Sin}[\theta]^2\ \mathrm{Sin}[\phi]^2}{(dG+(t-tP)\ \mathrm{Cos}[\theta])^2}\right)}\ +$$

$$\mathrm{Cos}[\theta]\ \sqrt{\left(dG^2+(t-tP)^2+2\ dG\ (t-tP)\ \mathrm{Cos}[\theta]\right)\ \left(1+\frac{(t-tP)^2\ \mathrm{Sin}[\theta]^2\ \mathrm{Sin}[\phi]^2}{(dG+(t-tP)\ \mathrm{Cos}[\theta])^2}\right)}$$

$$\Big((t-tP)^2\ (dG+(t-tP)\ \mathrm{Cos}[\theta])^2\ \mathrm{Cos}[2\ \phi]\ \mathrm{Sin}[\theta]^2-$$

$$\frac{1}{4}\ (t-tP)^4\ \mathrm{Sin}[\theta]^4\ \mathrm{Sin}[2\ \phi]^2\Big)\Big)\Big)\Big/$$

$$\Big(\left((dG+(t-tP)\ \mathrm{Cos}[\theta])^2+(t-tP)^2\ \mathrm{Cos}[\phi]^2\ \mathrm{Sin}[\theta]^2\right)$$

$$\left((dG+(t-tP)\ \mathrm{Cos}[\theta])^2+(t-tP)^2\ \mathrm{Sin}[\theta]^2\ \mathrm{Sin}[\phi]^2\right)$$

$$\sqrt{\left(dG^2+(t-tP)^2+2\ dG\ (t-tP)\ \mathrm{Cos}[\theta]\right)^2\ \left(1+\frac{(t-tP)^2\ \mathrm{Sin}[\theta]^2\ \mathrm{Sin}[\phi]^2}{(dG+(t-tP)\ \mathrm{Cos}[\theta])^2}\right)}\ \Big)$$

In[ ]:=

$$
\begin{aligned}
&\mathrm{FullSimplify}\Big[\Big(\mathrm{Cos}[\phi]\,\mathrm{Sin}[\theta]\,\left((\mathrm{dG}+(t-\mathrm{tP})\,\mathrm{Cos}[\theta])^2+(t-\mathrm{tP})^2\,\mathrm{Sin}[\theta]^2\,\mathrm{Sin}[\phi]^2\right)\\
&\quad\Big((t-\mathrm{tP})\,\mathrm{Cos}[\theta]\,(\mathrm{dG}+(t-\mathrm{tP})\,\mathrm{Cos}[\theta])\,\sqrt{\mathrm{dG}^2+(t-\mathrm{tP})^2+2\,\mathrm{dG}\,(t-\mathrm{tP})\,\mathrm{Cos}[\theta]}\;\mathrm{Cos}[\phi]\,\mathrm{Sin}[\theta]+\\
&\quad\;(t-\mathrm{tP})^2\,\sqrt{\mathrm{dG}^2+(t-\mathrm{tP})^2+2\,\mathrm{dG}\,(t-\mathrm{tP})\,\mathrm{Cos}[\theta]}\;\mathrm{Cos}[\phi]\,\mathrm{Sin}[\theta]^3\,\mathrm{Sin}[\phi]^2+(\mathrm{dG}+(t-\mathrm{tP})\,\mathrm{Cos}[\theta])^2\,\mathrm{Cos}[\phi]\\
&\quad\;\mathrm{Sin}[\theta]\,\sqrt{\left(\mathrm{dG}^2+(t-\mathrm{tP})^2+2\,\mathrm{dG}\,(t-\mathrm{tP})\,\mathrm{Cos}[\theta]\right)\left(1+\frac{(t-\mathrm{tP})^2\,\mathrm{Sin}[\theta]^2\,\mathrm{Sin}[\phi]^2}{(\mathrm{dG}+(t-\mathrm{tP})\,\mathrm{Cos}[\theta])^2}\right)}\Big)+\mathrm{Sin}[\theta]\,\mathrm{Sin}[\phi]\\
&\quad\Big((t-\mathrm{tP})^2\,\sqrt{\mathrm{dG}^2+(t-\mathrm{tP})^2+2\,\mathrm{dG}\,(t-\mathrm{tP})\,\mathrm{Cos}[\theta]}\;\mathrm{Cos}[\phi]^2\,\mathrm{Sin}[\theta]^3\,\mathrm{Sin}[\phi]\,\left((\mathrm{dG}+(t-\mathrm{tP})\,\mathrm{Cos}[\theta])^2+(t-\mathrm{tP})^2\,\mathrm{Sin}[\theta]^2\,\mathrm{Sin}[\phi]^2\right)+\\
&\quad\;(t-\mathrm{tP})\,\mathrm{Cos}[\theta]\,(\mathrm{dG}+(t-\mathrm{tP})\,\mathrm{Cos}[\theta])\,\mathrm{Sin}[\theta]\,\left((\mathrm{dG}+(t-\mathrm{tP})\,\mathrm{Cos}[\theta])^2+2\,(t-\mathrm{tP})^2\,\mathrm{Cos}[\phi]^2\,\mathrm{Sin}[\theta]^2\right)\\
&\quad\;\mathrm{Sin}[\phi]\,\sqrt{\left(\mathrm{dG}^2+(t-\mathrm{tP})^2+2\,\mathrm{dG}\,(t-\mathrm{tP})\,\mathrm{Cos}[\theta]\right)\left(1+\frac{(t-\mathrm{tP})^2\,\mathrm{Sin}[\theta]^2\,\mathrm{Sin}[\phi]^2}{(\mathrm{dG}+(t-\mathrm{tP})\,\mathrm{Cos}[\theta])^2}\right)}-\\
&\quad\;\mathrm{Sin}[\theta]\,\mathrm{Sin}[\phi]\,\sqrt{\left(\mathrm{dG}^2+(t-\mathrm{tP})^2+2\,\mathrm{dG}\,(t-\mathrm{tP})\,\mathrm{Cos}[\theta]\right)\left(1+\frac{(t-\mathrm{tP})^2\,\mathrm{Sin}[\theta]^2\,\mathrm{Sin}[\phi]^2}{(\mathrm{dG}+(t-\mathrm{tP})\,\mathrm{Cos}[\theta])^2}\right)}\\
&\quad\;\left((\mathrm{dG}+(t-\mathrm{tP})\,\mathrm{Cos}[\theta])^4+(t-\mathrm{tP})^2\,(\mathrm{dG}+(t-\mathrm{tP})\,\mathrm{Cos}[\theta])^2\,\mathrm{Cos}[\phi]^2\,\mathrm{Sin}[\theta]^2-(t-\mathrm{tP})^4\,\mathrm{Cos}[\phi]^2\,\mathrm{Sin}[\theta]^4\,\mathrm{Sin}[\phi]^2\right)\Big)+\\
&\quad\mathrm{Cos}[\theta]\,\Big((t-\mathrm{tP})\,(\mathrm{dG}+(t-\mathrm{tP})\,\mathrm{Cos}[\theta])\,\sqrt{\mathrm{dG}^2+(t-\mathrm{tP})^2+2\,\mathrm{dG}\,(t-\mathrm{tP})\,\mathrm{Cos}[\theta]}\;\mathrm{Cos}[\phi]^2\,\mathrm{Sin}[\theta]^2\\
&\quad\;\left((\mathrm{dG}+(t-\mathrm{tP})\,\mathrm{Cos}[\theta])^2+(t-\mathrm{tP})^2\,\mathrm{Sin}[\theta]^2\,\mathrm{Sin}[\phi]^2\right)+\\
&\quad\;(t-\mathrm{tP})\,(\mathrm{dG}+(t-\mathrm{tP})\,\mathrm{Cos}[\theta])\,\mathrm{Sin}[\theta]^2\,\left((\mathrm{dG}+(t-\mathrm{tP})\,\mathrm{Cos}[\theta])^2+2\,(t-\mathrm{tP})^2\,\mathrm{Cos}[\phi]^2\,\mathrm{Sin}[\theta]^2\right)\\
&\quad\;\mathrm{Sin}[\phi]^2\,\sqrt{\left(\mathrm{dG}^2+(t-\mathrm{tP})^2+2\,\mathrm{dG}\,(t-\mathrm{tP})\,\mathrm{Cos}[\theta]\right)\left(1+\frac{(t-\mathrm{tP})^2\,\mathrm{Sin}[\theta]^2\,\mathrm{Sin}[\phi]^2}{(\mathrm{dG}+(t-\mathrm{tP})\,\mathrm{Cos}[\theta])^2}\right)}+\\
&\quad\;\mathrm{Cos}[\theta]\,\sqrt{\left(\mathrm{dG}^2+(t-\mathrm{tP})^2+2\,\mathrm{dG}\,(t-\mathrm{tP})\,\mathrm{Cos}[\theta]\right)\left(1+\frac{(t-\mathrm{tP})^2\,\mathrm{Sin}[\theta]^2\,\mathrm{Sin}[\phi]^2}{(\mathrm{dG}+(t-\mathrm{tP})\,\mathrm{Cos}[\theta])^2}\right)}\\
&\quad\;\left((t-\mathrm{tP})^2\,(\mathrm{dG}+(t-\mathrm{tP})\,\mathrm{Cos}[\theta])^2\,\mathrm{Cos}[2\,\phi]\,\mathrm{Sin}[\theta]^2-\frac{1}{4}\,(t-\mathrm{tP})^4\,\mathrm{Sin}[\theta]^4\,\mathrm{Sin}[2\,\phi]^2\right)\Big)\Big)\Big/\\
&\Big(\left((\mathrm{dG}+(t-\mathrm{tP})\,\mathrm{Cos}[\theta])^2+(t-\mathrm{tP})^2\,\mathrm{Cos}[\phi]^2\,\mathrm{Sin}[\theta]^2\right)\left((\mathrm{dG}+(t-\mathrm{tP})\,\mathrm{Cos}[\theta])^2+(t-\mathrm{tP})^2\,\mathrm{Sin}[\theta]^2\,\mathrm{Sin}[\phi]^2\right)\\
&\quad\sqrt{\left(\mathrm{dG}^2+(t-\mathrm{tP})^2+2\,\mathrm{dG}\,(t-\mathrm{tP})\,\mathrm{Cos}[\theta]\right)^2\left(1+\frac{(t-\mathrm{tP})^2\,\mathrm{Sin}[\theta]^2\,\mathrm{Sin}[\phi]^2}{(\mathrm{dG}+(t-\mathrm{tP})\,\mathrm{Cos}[\theta])^2}\right)}\Big)\Big]
\end{aligned}
$$

```
In[•]:= Zint1[t_] := (Cos[ϕ] Sin[θ] ((dG + (t - tP) Cos[θ])^2 + (t - tP)^2 Sin[θ]^2 Sin[ϕ]^2)
        ((t - tP) Cos[θ] (dG + (t - tP) Cos[θ]) Sqrt[dG^2 + (t - tP)^2 + 2 dG (t - tP) Cos[θ]] Cos[ϕ] Sin[θ] +
          (t - tP)^2 Sqrt[dG^2 + (t - tP)^2 + 2 dG (t - tP) Cos[θ]] Cos[ϕ] Sin[θ]^3 Sin[ϕ]^2 +
          (dG + (t - tP) Cos[θ])^2 Cos[ϕ] Sin[θ] Sqrt[((dG^2 + (t - tP)^2 + 2 dG (t - tP) Cos[θ]) (1 + ((t - tP)^2 Sin[θ]^2 Sin[ϕ]^2)/(dG + (t - tP) Cos[θ])^2))]) +
       Sin[θ] Sin[ϕ] ((t - tP)^2 Sqrt[dG^2 + (t - tP)^2 + 2 dG (t - tP) Cos[θ]] Cos[ϕ]^2 Sin[θ]^3 Sin[ϕ]
          ((dG + (t - tP) Cos[θ])^2 + (t - tP)^2 Sin[θ]^2 Sin[ϕ]^2) +
          (t - tP) Cos[θ] (dG + (t - tP) Cos[θ]) Sin[θ] ((dG + (t - tP) Cos[θ])^2 + 2 (t - tP)^2 Cos[ϕ]^2 Sin[θ]^2)
          Sin[ϕ] Sqrt[((dG^2 + (t - tP)^2 + 2 dG (t - tP) Cos[θ]) (1 + ((t - tP)^2 Sin[θ]^2 Sin[ϕ]^2)/(dG + (t - tP) Cos[θ])^2))] -
          Sin[θ] Sin[ϕ] Sqrt[((dG^2 + (t - tP)^2 + 2 dG (t - tP) Cos[θ]) (1 + ((t - tP)^2 Sin[θ]^2 Sin[ϕ]^2)/(dG + (t - tP) Cos[θ])^2))]
          ((dG + (t - tP) Cos[θ])^4 + (t - tP)^2 (dG + (t - tP) Cos[θ])^2 Cos[ϕ]^2 Sin[θ]^2 - (t - tP)^4 Cos[ϕ]^2 Sin[θ]^4 Sin[ϕ]^2)) +
       Cos[θ] ((t - tP) (dG + (t - tP) Cos[θ]) Sqrt[dG^2 + (t - tP)^2 + 2 dG (t - tP) Cos[θ]] Cos[ϕ]^2 Sin[θ]^2
          ((dG + (t - tP) Cos[θ])^2 + (t - tP)^2 Sin[θ]^2 Sin[ϕ]^2) +
          (t - tP) (dG + (t - tP) Cos[θ]) Sin[θ]^2 ((dG + (t - tP) Cos[θ])^2 + 2 (t - tP)^2 Cos[ϕ]^2 Sin[θ]^2)
          Sin[ϕ]^2 Sqrt[((dG^2 + (t - tP)^2 + 2 dG (t - tP) Cos[θ]) (1 + ((t - tP)^2 Sin[θ]^2 Sin[ϕ]^2)/(dG + (t - tP) Cos[θ])^2))] +
          Cos[θ] Sqrt[((dG^2 + (t - tP)^2 + 2 dG (t - tP) Cos[θ]) (1 + ((t - tP)^2 Sin[θ]^2 Sin[ϕ]^2)/(dG + (t - tP) Cos[θ])^2))]
          ((t - tP)^2 (dG + (t - tP) Cos[θ])^2 Cos[2 ϕ] Sin[θ]^2 - 1/4 (t - tP)^4 Sin[θ]^4 Sin[2 ϕ]^2))) /
     (((dG + (t - tP) Cos[θ])^2 + (t - tP)^2 Cos[ϕ]^2 Sin[θ]^2) ((dG + (t - tP) Cos[θ])^2 + (t - tP)^2 Sin[θ]^2 Sin[ϕ]^2)
       Sqrt[((dG^2 + (t - tP)^2 + 2 dG (t - tP) Cos[θ])^2 (1 + ((t - tP)^2 Sin[θ]^2 Sin[ϕ]^2)/(dG + (t - tP) Cos[θ])^2))]);

In[•]:= Simplify[Zint1[Tmin[tG, tP, dG, θ]] - Zint1[tP]]

In[•]:= Z3d1[θ_, dG_, tP_, hM_, ϕ_] :=
  -(hM * dG)/2 (1/32 Sin[θ]^2 (-(32 Cos[2 ϕ])/dG + (Abs[-2 dG (dG + tP) + (2 dG^2 + 2 dG tP + tP^2) Cos[θ]]
        (16 (dG + tP - dG Cos[θ])^2 Cos[ϕ]^2 ((dG + (tP (2 dG + tP) Cos[θ])/(-2 (dG + tP) + 2 dG Cos[θ]))^2 +
            (tP^2 (2 dG + tP)^2 Sin[θ]^2 Sin[ϕ]^2)/(4 (dG + tP - dG Cos[θ])^2)) (tP (2 dG + tP) Cos[θ]
            (-2 dG (dG + tP) + (2 dG^2 + 2 dG tP + tP^2) Cos[θ]) + tP^2 (2 dG + tP)^2
            Sin[θ]^2 Sin[ϕ]^2 + ((-2 dG (dG + tP) + (2 dG^2 + 2 dG tP + tP^2) Cos[θ])^2
             Sqrt[(4 dG^2 (dG + tP)^2 - 4 dG (2 dG^3 + 4 dG^2 tP + 3 dG tP^2 + tP^3) Cos[θ] +
                 (2 dG^2 + 2 dG tP + tP^2)^2 Cos[θ]^2 + tP^2 (2 dG + tP)^2 Sin[θ]^2 Sin[ϕ]^2)]) /
          Abs[-2 dG (dG + tP) + (2 dG^2 + 2 dG tP + tP^2) Cos[θ]]) + 4 Sin[ϕ]^2
```

$$\Bigg(\Big(tP\ (2\ dG + tP)\ Cos[\theta]\ \big(-2\ dG\ (dG + tP) + \big(2\ dG^2 + 2\ dG\ tP + tP^2\big)\ Cos[\theta]\big)$$
$$\big(4\ dG^2\ (dG + tP)^2 - 4\ dG\ \big(2\ dG^3 + 4\ dG^2\ tP + 3\ dG\ tP^2 + tP^3\big)\ Cos[\theta] +$$
$$\big(2\ dG^2 + 2\ dG\ tP + tP^2\big)^2\ Cos[\theta]^2 + 2\ tP^2\ (2\ dG + tP)^2\ Cos[\phi]^2\ Sin[\theta]^2\big)$$
$$\sqrt{\big(4\ dG^2\ (dG + tP)^2 - 4\ dG\ \big(2\ dG^3 + 4\ dG^2\ tP + 3\ dG\ tP^2 + tP^3\big)\ Cos[\theta] +}$$
$$\big(2\ dG^2 + 2\ dG\ tP + tP^2\big)^2\ Cos[\theta]^2 + tP^2\ (2\ dG + tP)^2\ Sin[\theta]^2\ Sin[\phi]^2\big)\Big)\Big/$$
$$Abs\big[-2\ dG\ (dG + tP) + \big(2\ dG^2 + 2\ dG\ tP + tP^2\big)\ Cos[\theta]\big] + tP^2\ (2\ dG + tP)^2$$
$$Cos[\phi]^2\ Sin[\theta]^2\ \big(4\ dG^2\ (dG + tP)^2 - 4\ dG\ \big(2\ dG^3 + 4\ dG^2\ tP + 3\ dG\ tP^2 + tP^3\big)\ Cos[$$
$$\theta] + \big(2\ dG^2 + 2\ dG\ tP + tP^2\big)^2\ Cos[\theta]^2 + tP^2\ (2\ dG + tP)^2\ Sin[\theta]^2\ Sin[\phi]^2\big) -$$
$$\Big(16\ (dG + tP - dG\ Cos[\theta])^4\ \sqrt{\big(4\ dG^2\ (dG + tP)^2 - 4\ dG\ \big(2\ dG^3 + 4\ dG^2\ tP +}$$
$$3\ dG\ tP^2 + tP^3\big)\ Cos[\theta] + \big(2\ dG^2 + 2\ dG\ tP + tP^2\big)^2\ Cos[\theta]^2 + tP^2$$
$$(2\ dG + tP)^2\ Sin[\theta]^2\ Sin[\phi]^2\big)\ \Bigg(\Bigg(dG + \frac{tP\ (2\ dG + tP)\ Cos[\theta]}{-2\ (dG + tP) + 2\ dG\ Cos[\theta]}\Bigg)^4 +$$
$$\big(tP^2\ (2\ dG + tP)^2\ \big(-2\ dG\ (dG + tP) + \big(2\ dG^2 + 2\ dG\ tP + tP^2\big)\ Cos[\theta]\big)^2$$
$$Cos[\phi]^2\ Sin[\theta]^2\big)\Big/\ \big(16\ (dG + tP - dG\ Cos[\theta])^4\big) -$$
$$\frac{tP^4\ (2\ dG + tP)^4\ Cos[\phi]^2\ Sin[\theta]^4\ Sin[\phi]^2}{16\ (dG + tP - dG\ Cos[\theta])^4}\Bigg)\Bigg)\Big/\ Abs\big[$$
$$-2\ dG\ (dG + tP) + \big(2\ dG^2 + 2\ dG\ tP + tP^2\big)\ Cos[\theta]\big]\Big) + tP\ (2\ dG + tP)\ Cos[\theta]$$
$$\Big(\Big(4\ \big(-2\ dG\ (dG + tP) + \big(2\ dG^2 + 2\ dG\ tP + tP^2\big)\ Cos[\theta]\big)\ \big(4\ dG^2\ (dG + tP)^2 -$$
$$4\ dG\ \big(2\ dG^3 + 4\ dG^2\ tP + 3\ dG\ tP^2 + tP^3\big)\ Cos[\theta] + \big(2\ dG^2 + 2\ dG\ tP + tP^2\big)^2$$
$$Cos[\theta]^2 + 2\ tP^2\ (2\ dG + tP)^2\ Cos[\phi]^2\ Sin[\theta]^2\big)\ Sin[\phi]^2$$
$$\sqrt{\big(4\ dG^2\ (dG + tP)^2 - 4\ dG\ \big(2\ dG^3 + 4\ dG^2\ tP + 3\ dG\ tP^2 + tP^3\big)\ Cos[\theta] +}$$
$$\big(2\ dG^2 + 2\ dG\ tP + tP^2\big)^2\ Cos[\theta]^2 + tP^2\ (2\ dG + tP)^2\ Sin[\theta]^2\ Sin[\phi]^2\big)\Big)\Big/$$
$$Abs\big[-2\ dG\ (dG + tP) + \big(2\ dG^2 + 2\ dG\ tP + tP^2\big)\ Cos[\theta]\big] + 4$$
$$\big(-2\ dG\ (dG + tP) + \big(2\ dG^2 + 2\ dG\ tP + tP^2\big)\ Cos[\theta]\big)\ Cos[\phi]^2\ \big(4\ dG^2\ (dG + tP)^2 -$$
$$4\ dG\ \big(2\ dG^3 + 4\ dG^2\ tP + 3\ dG\ tP^2 + tP^3\big)\ Cos[\theta] + \big(2\ dG^2 + 2\ dG\ tP + tP^2\big)^2$$
$$Cos[\theta]^2 + tP^2\ (2\ dG + tP)^2\ Sin[\theta]^2\ Sin[\phi]^2\big) + \big(tP\ (2\ dG + tP)\ Cos[\theta]$$
$$\sqrt{\big(4\ dG^2\ (dG + tP)^2 - 4\ dG\ \big(2\ dG^3 + 4\ dG^2\ tP + 3\ dG\ tP^2 + tP^3\big)\ Cos[\theta] +}$$
$$\big(2\ dG^2 + 2\ dG\ tP + tP^2\big)^2\ Cos[\theta]^2 + tP^2\ (2\ dG + tP)^2\ Sin[\theta]^2\ Sin[\phi]^2\big)$$
$$\big(4\ \big(-2\ dG\ (dG + tP) + \big(2\ dG^2 + 2\ dG\ tP + tP^2\big)\ Cos[\theta]\big)^2\ Cos[2\ \phi] -$$
$$tP^2\ (2\ dG + tP)^2\ Sin[\theta]^2\ Sin[2\ \phi]^2\big)\big)\Big/\ Abs\big[$$

$$-2\,dG\,(dG+tP)+\left(2\,dG^2+2\,dG\,tP+tP^2\right)\mathrm{Cos}[\theta]\Big]\Big)\Big)\Big)\Big/$$

$$\Bigg((dG+tP-dG\,\mathrm{Cos}[\theta])^3\left(2\,dG^2+2\,dG\,tP+tP^2-2\,dG\,(dG+tP)\,\mathrm{Cos}[\theta]\right)$$

$$\left(\left(dG+\frac{tP\,(2\,dG+tP)\,\mathrm{Cos}[\theta]}{-2\,(dG+tP)+2\,dG\,\mathrm{Cos}[\theta]}\right)^2+\frac{tP^2\,(2\,dG+tP)^2\,\mathrm{Cos}[\phi]^2\,\mathrm{Sin}[\theta]^2}{4\,(dG+tP-dG\,\mathrm{Cos}[\theta])^2}\right)$$

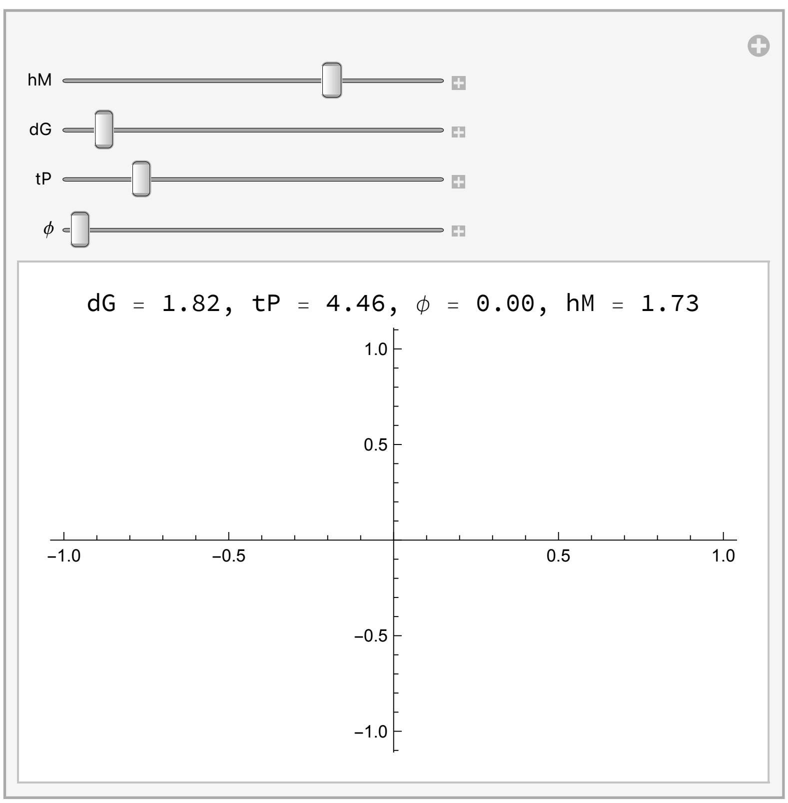
$$\sqrt{\Big(4\,dG^2\,(dG+tP)^2-4\,dG\left(2\,dG^3+4\,dG^2\,tP+3\,dG\,tP^2+tP^3\right)\mathrm{Cos}[\theta]+\left(2\,dG^2+2\,dG\,tP+tP^2\right)^2\mathrm{Cos}[\theta]^2+tP^2\,(2\,dG+tP)^2\,\mathrm{Sin}[\theta]^2\,\mathrm{Sin}[\phi]^2\Big)}$$

$$\left(\left(dG+\frac{tP\,(2\,dG+tP)\,\mathrm{Cos}[\theta]}{-2\,(dG+tP)+2\,dG\,\mathrm{Cos}[\theta]}\right)^2+\frac{tP^2\,(2\,dG+tP)^2\,\mathrm{Sin}[\theta]^2\,\mathrm{Sin}[\phi]^2}{4\,(dG+tP-dG\,\mathrm{Cos}[\theta])^2}\right)\Bigg)\Bigg)\Bigg)$$

In[∘]:=
```
Z3d1[0.5, 0.5, 0.3, 1, 0]
```
Out[∘]=

-0.53878

In[∘]:=
```
Manipulate[Labeled[Plot[Z3d1[θ * π, dG, tP, hM, ϕ], {θ, 0, 1}, PlotRange → All],
   Row[{"dG = ", NumberForm[dG, {3, 2}], ", tP = ", NumberForm[tP, {3, 2}],
     ", ϕ = ", NumberForm[ϕ, {3, 2}], ", hM = ", NumberForm[hM, {3, 2}]}], Top],
  {hM, 1, 2}, {dG, 0.1, 25}, {tP, 0.01, 25}, {ϕ, 0, 2 π}]
```
Out[∘]=



Second case, true

In[ ]:=
```
Simplify[v[θ, ϕ].Transpose[v[θ, ϕ].hp3d2simp]]
```

In[ ]:=
```
FullSimplify[(-2 (dG + (t - tP) Cos[θ]) Cos[ϕ] Sin[θ] ((dG + (t - tP) Cos[θ])^2 + (t - tP)^2 Sin[θ]^2 Sin[ϕ]^2)
    ((t - tP) Cos[θ] (dG + (t - tP) Cos[θ]) Sqrt[dG^2 + (t - tP)^2 + 2 dG (t - tP) Cos[θ]] Cos[ϕ] Sin[θ] +
      (t - tP)^2 Sqrt[dG^2 + (t - tP)^2 + 2 dG (t - tP) Cos[θ]] Cos[ϕ] Sin[θ]^3 Sin[ϕ]^2 -
      (dG + (t - tP) Cos[θ])^2 Cos[ϕ] Sin[θ] Sqrt[(dG^2 + (t - tP)^2 + 2 dG (t - tP) Cos[θ]) (1 + ((t - tP)^2 Sin[θ]^2 Sin[ϕ]^2)/(dG + (t - tP) Cos[θ])^2)]) -
   2 Cos[θ] (dG + (t - tP) Cos[θ]) ((t - tP) (dG + (t - tP) Cos[θ]) Sqrt[dG^2 + (t - tP)^2 + 2 dG (t - tP) Cos[θ]]
       Cos[ϕ]^2 Sin[θ]^2 ((dG + (t - tP) Cos[θ])^2 + (t - tP)^2 Sin[θ]^2 Sin[ϕ]^2) -
      (t - tP) (dG + (t - tP) Cos[θ]) Sin[θ]^2 ((dG + (t - tP) Cos[θ])^2 + 2 (t - tP)^2 Cos[ϕ]^2 Sin[θ]^2)
       Sin[ϕ]^2 Sqrt[(dG^2 + (t - tP)^2 + 2 dG (t - tP) Cos[θ]) (1 + ((t - tP)^2 Sin[θ]^2 Sin[ϕ]^2)/(dG + (t - tP) Cos[θ])^2)] -
      Cos[θ] Sqrt[(dG^2 + (t - tP)^2 + 2 dG (t - tP) Cos[θ]) (1 + ((t - tP)^2 Sin[θ]^2 Sin[ϕ]^2)/(dG + (t - tP) Cos[θ])^2)]
       ((t - tP)^2 (dG + (t - tP) Cos[θ])^2 Cos[2 ϕ] Sin[θ]^2 - 1/4 (t - tP)^4 Sin[θ]^4 Sin[2 ϕ]^2)) +
   Sin[θ]^2 Sin[ϕ]^2 (2 (t - tP) Cos[θ] (dG + (t - tP) Cos[θ])^2 ((dG + (t - tP) Cos[θ])^2 + 2 (t - tP)^2 Cos[ϕ]^2 Sin[θ]^2)
       Sqrt[(dG^2 + (t - tP)^2 + 2 dG (t - tP) Cos[θ]) (1 + ((t - tP)^2 Sin[θ]^2 Sin[ϕ]^2)/(dG + (t - tP) Cos[θ])^2)] -
      2 (dG + (t - tP) Cos[θ]) Sqrt[(dG^2 + (t - tP)^2 + 2 dG (t - tP) Cos[θ]) (1 + ((t - tP)^2 Sin[θ]^2 Sin[ϕ]^2)/(dG + (t - tP) Cos[θ])^2)]
       ((dG + (t - tP) Cos[θ])^4 + (t - tP)^2 (dG + (t - tP) Cos[θ])^2 Cos[ϕ]^2 Sin[θ]^2 - (t - tP)^4 Cos[ϕ]^2 Sin[θ]^4 Sin[ϕ]^2) +
      (t - tP) Sqrt[dG^2 + (t - tP)^2 + 2 dG (t - tP) Cos[θ]] Cos[ϕ] Sin[θ] ((dG + (t - tP) Cos[θ])^2 + (t - tP)^2 Cos[ϕ]^2 Sin[θ]^2)
       ((dG + (t - tP) Cos[θ])^2 + (t - tP)^2 Sin[θ]^2 Sin[ϕ]^2) Sin[2 ArcTan[((-t + tP) Cos[ϕ] Sin[θ])/(dG + (t - tP) Cos[θ])]])) /
  (2 (dG + (t - tP) Cos[θ]) ((dG + (t - tP) Cos[θ])^2 + (t - tP)^2 Cos[ϕ]^2 Sin[θ]^2)
    ((dG + (t - tP) Cos[θ])^2 + (t - tP)^2 Sin[θ]^2 Sin[ϕ]^2)
    Sqrt[(dG^2 + (t - tP)^2 + 2 dG (t - tP) Cos[θ])^2 (1 + ((t - tP)^2 Sin[θ]^2 Sin[ϕ]^2)/(dG + (t - tP) Cos[θ])^2)])]
```

In[ ]:=
```
Zint2[t_] :=
  (-4 Cos[ϕ]^2 Sin[θ]^2 ((dG + (t - tP) Cos[θ])^2 + (t - tP)^2 Sin[θ]^2 Sin[ϕ]^2) ((t - tP) Cos[θ]
       (dG + (t - tP) Cos[θ]) Sqrt[dG^2 + (t - tP)^2 + 2 dG (t - tP) Cos[θ]] + (t - tP)^2
       Sqrt[dG^2 + (t - tP)^2 + 2 dG (t - tP) Cos[θ]] Sin[θ]^2 Sin[ϕ]^2 - (dG + (t - tP) Cos[θ])^2
```

$$\sqrt{\left(dG^2+(t-tP)^2+2\,dG\,(t-tP)\,\mathrm{Cos}[\theta]\right)\left(1+\frac{(t-tP)^2\,\mathrm{Sin}[\theta]^2\,\mathrm{Sin}[\phi]^2}{(dG+(t-tP)\,\mathrm{Cos}[\theta])^2}\right)}\Bigg)+$$

$$\mathrm{Sin}[\theta]^2\,\mathrm{Sin}[\phi]^2\,\Bigg(-(t-tP)^2\,\sqrt{dG^2+(t-tP)^2+2\,dG\,(t-tP)\,\mathrm{Cos}[\theta]}$$

$$\mathrm{Cos}[\phi]^2\,\mathrm{Sin}[\theta]^2\,\Big(4\,dG^2+3\,(t-tP)^2+8\,dG\,(t-tP)\,\mathrm{Cos}[\theta]+$$

$$(t-tP)^2\,\left(\mathrm{Cos}[2\,\theta]-2\,\mathrm{Cos}[2\,\phi]\,\mathrm{Sin}[\theta]^2\right)\Big)-$$

$$4\,\sqrt{\left(dG^2+(t-tP)^2+2\,dG\,(t-tP)\,\mathrm{Cos}[\theta]\right)\left(1+\frac{(t-tP)^2\,\mathrm{Sin}[\theta]^2\,\mathrm{Sin}[\phi]^2}{(dG+(t-tP)\,\mathrm{Cos}[\theta])^2}\right)}$$

$$\Big(dG^4+(t-tP)\,\Big(3\,dG^3\,\mathrm{Cos}[\theta]+dG\,(t-tP)^2\,\mathrm{Cos}[\theta]^3+$$

$$(t-tP)\,\mathrm{Cos}[\theta]^2\,\left(3\,dG^2-(t-tP)^2\,\mathrm{Cos}[\phi]^2\,\mathrm{Sin}[\theta]^2\right)+$$

$$(t-tP)\,\mathrm{Cos}[\phi]^2\,\mathrm{Sin}[\theta]^2\,\left(dG^2-(t-tP)^2\,\mathrm{Sin}[\theta]^2\,\mathrm{Sin}[\phi]^2\right)\Big)\Big)\Bigg)-$$

$$4\,\mathrm{Cos}[\theta]\,\Bigg((t-tP)\,(dG+(t-tP)\,\mathrm{Cos}[\theta])\,\sqrt{dG^2+(t-tP)^2+2\,dG\,(t-tP)\,\mathrm{Cos}[\theta]}$$

$$\mathrm{Cos}[\phi]^2\,\mathrm{Sin}[\theta]^2\,\left((dG+(t-tP)\,\mathrm{Cos}[\theta])^2+(t-tP)^2\,\mathrm{Sin}[\theta]^2\,\mathrm{Sin}[\phi]^2\right)-$$

$$(t-tP)\,(dG+(t-tP)\,\mathrm{Cos}[\theta])\,\mathrm{Sin}[\theta]^2$$

$$\left((dG+(t-tP)\,\mathrm{Cos}[\theta])^2+2\,(t-tP)^2\,\mathrm{Cos}[\phi]^2\,\mathrm{Sin}[\theta]^2\right)\,\mathrm{Sin}[\phi]^2$$

$$\sqrt{\left(dG^2+(t-tP)^2+2\,dG\,(t-tP)\,\mathrm{Cos}[\theta]\right)\left(1+\frac{(t-tP)^2\,\mathrm{Sin}[\theta]^2\,\mathrm{Sin}[\phi]^2}{(dG+(t-tP)\,\mathrm{Cos}[\theta])^2}\right)}-$$

$$\mathrm{Cos}[\theta]\,\sqrt{\left(dG^2+(t-tP)^2+2\,dG\,(t-tP)\,\mathrm{Cos}[\theta]\right)\left(1+\frac{(t-tP)^2\,\mathrm{Sin}[\theta]^2\,\mathrm{Sin}[\phi]^2}{(dG+(t-tP)\,\mathrm{Cos}[\theta])^2}\right)}$$

$$\Big((t-tP)^2\,(dG+(t-tP)\,\mathrm{Cos}[\theta])^2\,\mathrm{Cos}[2\,\phi]\,\mathrm{Sin}[\theta]^2-$$

$$\frac{1}{4}\,(t-tP)^4\,\mathrm{Sin}[\theta]^4\,\mathrm{Sin}[2\,\phi]^2\Big)\Bigg)\Bigg)\Bigg/$$

$$\Bigg(4\,\left((dG+(t-tP)\,\mathrm{Cos}[\theta])^2+(t-tP)^2\,\mathrm{Cos}[\phi]^2\,\mathrm{Sin}[\theta]^2\right)$$

$$\left((dG+(t-tP)\,\mathrm{Cos}[\theta])^2+(t-tP)^2\,\mathrm{Sin}[\theta]^2\,\mathrm{Sin}[\phi]^2\right)$$

$$\sqrt{\left(dG^2+(t-tP)^2+2\,dG\,(t-tP)\,\mathrm{Cos}[\theta]\right)^2\left(1+\frac{(t-tP)^2\,\mathrm{Sin}[\theta]^2\,\mathrm{Sin}[\phi]^2}{(dG+(t-tP)\,\mathrm{Cos}[\theta])^2}\right)}\Bigg);$$

In[ ]:= `Simplify[Zint2[Tmin[tG, tP, dG, θ]] - Zint2[tP]]`

In[ ]:= `Z3d2[θ_, dG_, tP_, hM_, ϕ_] :=`

$$-\frac{hM*dG}{2}\,\Bigg(\frac{1}{32}\,\mathrm{Sin}[\theta]^2\,\Bigg(-\frac{32\,\mathrm{Cos}[2\,\phi]}{dG}+\Bigg(\mathrm{Abs}\left[-2\,dG\,(dG+tP)+\left(2\,dG^2+2\,dG\,tP+tP^2\right)\mathrm{Cos}[\theta]\right]$$

$$\left(-16\,(dG+tP-dG\,Cos[\theta])^2\,Cos[\phi]^2\left(\left(dG+\frac{tP\,(2\,dG+tP)\,Cos[\theta]}{-2\,(dG+tP)+2\,dG\,Cos[\theta]}\right)^2+\frac{tP^2\,(2\,dG+tP)^2\,Sin[\theta]^2\,Sin[\phi]^2}{4\,(dG+tP-dG\,Cos[\theta])^2}\right)\left(tP\,(2\,dG+tP)\,Cos[\theta]\right.\right.$$

$$\left(-2\,dG\,(dG+tP)+\left(2\,dG^2+2\,dG\,tP+tP^2\right)Cos[\theta]\right)+tP^2\,(2\,dG+tP)^2$$

$$Sin[\theta]^2\,Sin[\phi]^2-\left(\left(-2\,dG\,(dG+tP)+\left(2\,dG^2+2\,dG\,tP+tP^2\right)Cos[\theta]\right)^2\right.$$

$$\sqrt{\left(4\,dG^2\,(dG+tP)^2-4\,dG\left(2\,dG^3+4\,dG^2\,tP+3\,dG\,tP^2+tP^3\right)Cos[\theta]+\left(2\,dG^2+2\,dG\,tP+tP^2\right)^2Cos[\theta]^2+tP^2\,(2\,dG+tP)^2\,Sin[\theta]^2\,Sin[\phi]^2\right)}\Bigg)\Bigg/$$

$$\left.Abs\left[-2\,dG\,(dG+tP)+\left(2\,dG^2+2\,dG\,tP+tP^2\right)Cos[\theta]\right]\right)+Sin[\phi]^2$$

$$\left(-tP^2\,(2\,dG+tP)^2\,Cos[\phi]^2\,Sin[\theta]^2\left(16\,dG^4+32\,dG^3\,tP+28\,dG^2\,tP^2+12\,dG\,tP^3+\right.\right.$$

$$3\,tP^4-16\,dG\left(2\,dG^3+4\,dG^2\,tP+3\,dG\,tP^2+tP^3\right)Cos[\theta]+16\,dG^2\,(dG+tP)^2$$

$$Cos[\theta]^2+tP^2\,(2\,dG+tP)^2\,Cos[2\,\theta]-8\,dG^2\,tP^2\,Cos[2\,\phi]\,Sin[\theta]^2-8\,dG\,tP^3$$

$$\left.Cos[2\,\phi]\,Sin[\theta]^2-2\,tP^4\,Cos[2\,\phi]\,Sin[\theta]^2\right)-\left(64\,(dG+tP-dG\,Cos[\theta])^4\right.$$

$$\sqrt{\left(4\,dG^2\,(dG+tP)^2-4\,dG\left(2\,dG^3+4\,dG^2\,tP+3\,dG\,tP^2+tP^3\right)Cos[\theta]+\left(2\,dG^2+2\,dG\,tP+tP^2\right)^2Cos[\theta]^2+tP^2\,(2\,dG+tP)^2\,Sin[\theta]^2\,Sin[\phi]^2\right)}$$

$$\left(dG^4+\frac{1}{8\,(-2\,(dG+tP)+2\,dG\,Cos[\theta])}\,tP\,(2\,dG+tP)\left(24\,dG^3\,Cos[\theta]+\frac{2\,dG\,tP^2\,(2\,dG+tP)^2\,Cos[\theta]^3}{(dG+tP-dG\,Cos[\theta])^2}+\frac{1}{(dG+tP-dG\,Cos[\theta])^3}\right.\right.$$

$$tP\,(2\,dG+tP)\,Cos[\theta]^2\left(-12\,dG^2\,(dG+tP)^2+24\,dG^3\,(dG+tP)\,Cos[\theta]-12\,dG^4\,Cos[\theta]^2+tP^2\,(2\,dG+tP)^2\,Cos[\phi]^2\,Sin[\theta]^2\right)+$$

$$\frac{1}{(dG+tP-dG\,Cos[\theta])^3}\,tP\,(2\,dG+tP)\,Cos[\phi]^2\,Sin[\theta]^2$$

$$\left.\left.\left.\left(-4\,dG^2\,(dG+tP)^2+8\,dG^3\,(dG+tP)\,Cos[\theta]-4\,dG^4\,Cos[\theta]^2+tP^2\,(2\,dG+tP)^2\,Sin[\theta]^2\,Sin[\phi]^2\right)\right)\right)\right)\Big/\,Abs\Big[$$

$$\left.-2\,dG\,(dG+tP)+\left(2\,dG^2+2\,dG\,tP+tP^2\right)Cos[\theta]\Big]\right)+tP\,(2\,dG+tP)$$

$$Cos[\theta]\left(\left(4\left(-2\,dG\,(dG+tP)+\left(2\,dG^2+2\,dG\,tP+tP^2\right)Cos[\theta]\right)\left(4\,dG^2\,(dG+tP)^2-\right.\right.\right.$$

$$4\,dG\left(2\,dG^3+4\,dG^2\,tP+3\,dG\,tP^2+tP^3\right)Cos[\theta]+\left(2\,dG^2+2\,dG\,tP+tP^2\right)^2$$

$$\left.Cos[\theta]^2+2\,tP^2\,(2\,dG+tP)^2\,Cos[\phi]^2\,Sin[\theta]^2\right)Sin[\phi]^2$$

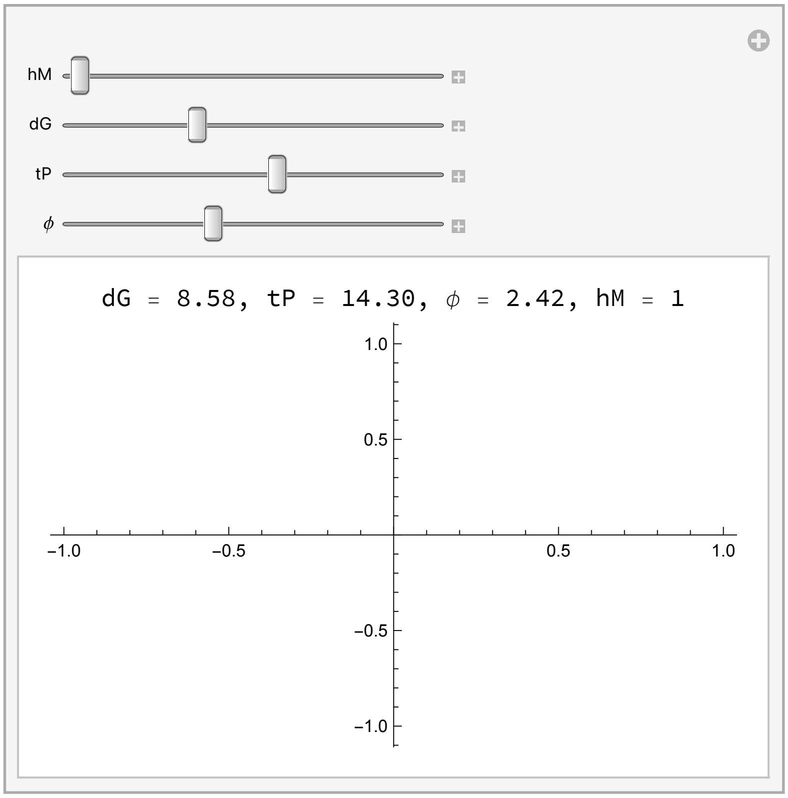
$$\sqrt{\left(4\,dG^2\,(dG+tP)^2-4\,dG\left(2\,dG^3+4\,dG^2\,tP+3\,dG\,tP^2+tP^3\right)Cos[\theta]+\left(2\,dG^2+2\,dG\,tP+tP^2\right)^2Cos[\theta]^2+tP^2\,(2\,dG+tP)^2\,Sin[\theta]^2\,Sin[\phi]^2\right)}\Bigg)\Bigg/$$

$$Abs\left[-2\,dG\,(dG+tP)+\left(2\,dG^2+2\,dG\,tP+tP^2\right)Cos[\theta]\right]-4$$

$$\left(-2\,dG\,(dG+tP)+\left(2\,dG^2+2\,dG\,tP+tP^2\right)Cos[\theta]\right)Cos[\phi]^2\left(4\,dG^2\,(dG+tP)^2-\right.$$
$$4\,dG\left(2\,dG^3+4\,dG^2\,tP+3\,dG\,tP^2+tP^3\right)Cos[\theta]+\left(2\,dG^2+2\,dG\,tP+tP^2\right)^2$$
$$\left.Cos[\theta]^2+tP^2\,(2\,dG+tP)^2\,Sin[\theta]^2\,Sin[\phi]^2\right)+\left(tP\,(2\,dG+tP)\,Cos[\theta]\right.$$
$$\sqrt{}\left(4\,dG^2\,(dG+tP)^2-4\,dG\left(2\,dG^3+4\,dG^2\,tP+3\,dG\,tP^2+tP^3\right)Cos[\theta]+\right.$$
$$\left.\left(2\,dG^2+2\,dG\,tP+tP^2\right)^2Cos[\theta]^2+tP^2\,(2\,dG+tP)^2\,Sin[\theta]^2\,Sin[\phi]^2\right)$$
$$\left(4\left(-2\,dG\,(dG+tP)+\left(2\,dG^2+2\,dG\,tP+tP^2\right)Cos[\theta]\right)^2Cos[2\,\phi]-\right.$$
$$\left.\left.tP^2\,(2\,dG+tP)^2\,Sin[\theta]^2\,Sin[2\,\phi]^2\right)\right)\Big/\,Abs\Big[$$
$$\left.\left.\left.-2\,dG\,(dG+tP)+\left(2\,dG^2+2\,dG\,tP+tP^2\right)Cos[\theta]\Big]\right)\right)\right)\Big/$$
$$\Big((dG+tP-dG\,Cos[\theta])^3\left(2\,dG^2+2\,dG\,tP+tP^2-2\,dG\,(dG+tP)\,Cos[\theta]\right)$$
$$\left(\left(dG+\frac{tP\,(2\,dG+tP)\,Cos[\theta]}{-2\,(dG+tP)+2\,dG\,Cos[\theta]}\right)^2+\right.$$
$$\left.\frac{tP^2\,(2\,dG+tP)^2\,Cos[\phi]^2\,Sin[\theta]^2}{4\,(dG+tP-dG\,Cos[\theta])^2}\right)$$
$$\sqrt{}\left(4\,dG^2\,(dG+tP)^2-4\,dG\left(2\,dG^3+4\,dG^2\,tP+3\,dG\,tP^2+tP^3\right)Cos[\theta]+\right.$$
$$\left.\left(2\,dG^2+2\,dG\,tP+tP^2\right)^2Cos[\theta]^2+tP^2\,(2\,dG+tP)^2\,Sin[\theta]^2\,Sin[\phi]^2\right)$$
$$\left.\left(\left(dG+\frac{tP\,(2\,dG+tP)\,Cos[\theta]}{-2\,(dG+tP)+2\,dG\,Cos[\theta]}\right)^2+\frac{tP^2\,(2\,dG+tP)^2\,Sin[\theta]^2\,Sin[\phi]^2}{4\,(dG+tP-dG\,Cos[\theta])^2}\right)\right)\Big)\Big)$$

In[ ]:=
```
Manipulate[Labeled[Plot[Z3d2[θ * π, dG, tP, hM, ϕ], {θ, 0, 1}, PlotRange → All],
  Row[{"dG = ", NumberForm[dG, {3, 2}], ", tP = ", NumberForm[tP, {3, 2}],
    ", ϕ = ", NumberForm[ϕ, {3, 2}], ", hM = ", NumberForm[hM, {3, 2}]}], Top],
 {hM, 1, 2}, {dG, 0.1, 25}, {tP, 0.01, 25}, {ϕ, 0, 2 π}]
```

Out[ ]=



In[ ]:=
```
Simplify[Zint1[Tmin[tG, tP, dG, θ]] - Zint2[tP]] (*If the condition is true*)
```

Out[ ]=

$$\frac{1}{32}\,\mathrm{Sin}[\theta]^2\left(-\frac{32\,\mathrm{Cos}[2\,\phi]}{dG}+\right.$$

$$\Big(\mathrm{Abs}\big[-2\,dG\,(dG+tP)+\left(2\,dG^2+2\,dG\,tP+tP^2\right)\mathrm{Cos}[\theta]\big]\,\Big(16\,(dG+tP-dG\,\mathrm{Cos}[\theta])^2\,\mathrm{Cos}[\phi]^2$$

$$\left(\left(dG+\frac{tP\,(2\,dG+tP)\,\mathrm{Cos}[\theta]}{-2\,(dG+tP)+2\,dG\,\mathrm{Cos}[\theta]}\right)^2+\frac{tP^2\,(2\,dG+tP)^2\,\mathrm{Sin}[\theta]^2\,\mathrm{Sin}[\phi]^2}{4\,(dG+tP-dG\,\mathrm{Cos}[\theta])^2}\right)$$

$$\Big(tP\,(2\,dG+tP)\,\mathrm{Cos}[\theta]\,\left(-2\,dG\,(dG+tP)+\left(2\,dG^2+2\,dG\,tP+tP^2\right)\mathrm{Cos}[\theta]\right)+tP^2\,(2\,dG+$$

$$tP)^2\,\mathrm{Sin}[\theta]^2\,\mathrm{Sin}[\phi]^2+\Big(\left(-2\,dG\,(dG+tP)+\left(2\,dG^2+2\,dG\,tP+tP^2\right)\mathrm{Cos}[\theta]\right)^2$$

$$\sqrt{}\Big(4\,dG^2\,(dG+tP)^2-4\,dG\,\left(2\,dG^3+4\,dG^2\,tP+3\,dG\,tP^2+tP^3\right)\mathrm{Cos}[\theta]+$$

$$\left(2\,dG^2+2\,dG\,tP+tP^2\right)^2\mathrm{Cos}[\theta]^2+tP^2\,(2\,dG+tP)^2\,\mathrm{Sin}[\theta]^2\,\mathrm{Sin}[\phi]^2\Big)\Big)\Big/$$

$$\mathrm{Abs}\big[-2\,dG\,(dG+tP)+\left(2\,dG^2+2\,dG\,tP+tP^2\right)\mathrm{Cos}[\theta]\big]\Big)+$$

$$4\,\text{Sin}[\phi]^2\,\Big(\Big(tP\,(2\,dG+tP)\,\text{Cos}[\theta]\,\big(-2\,dG\,(dG+tP)+(2\,dG^2+2\,dG\,tP+tP^2)\,\text{Cos}[\theta]\big)$$
$$\big(4\,dG^2\,(dG+tP)^2-4\,dG\,(2\,dG^3+4\,dG^2\,tP+3\,dG\,tP^2+tP^3)\,\text{Cos}[\theta]+$$
$$(2\,dG^2+2\,dG\,tP+tP^2)^2\,\text{Cos}[\theta]^2+2\,tP^2\,(2\,dG+tP)^2\,\text{Cos}[\phi]^2\,\text{Sin}[\theta]^2\big)$$
$$\surd\big(4\,dG^2\,(dG+tP)^2-4\,dG\,(2\,dG^3+4\,dG^2\,tP+3\,dG\,tP^2+tP^3)\,\text{Cos}[\theta]+$$
$$(2\,dG^2+2\,dG\,tP+tP^2)^2\,\text{Cos}[\theta]^2+tP^2\,(2\,dG+tP)^2\,\text{Sin}[\theta]^2\,\text{Sin}[\phi]^2\big)\big)\Big/$$
$$\text{Abs}\big[-2\,dG\,(dG+tP)+(2\,dG^2+2\,dG\,tP+tP^2)\,\text{Cos}[\theta]\big]+tP^2\,(2\,dG+tP)^2\,\text{Cos}[\phi]^2$$
$$\text{Sin}[\theta]^2\,\big(4\,dG^2\,(dG+tP)^2-4\,dG\,(2\,dG^3+4\,dG^2\,tP+3\,dG\,tP^2+tP^3)\,\text{Cos}[\theta]+$$
$$(2\,dG^2+2\,dG\,tP+tP^2)^2\,\text{Cos}[\theta]^2+tP^2\,(2\,dG+tP)^2\,\text{Sin}[\theta]^2\,\text{Sin}[\phi]^2\big)-$$
$$\Big(16\,(dG+tP-dG\,\text{Cos}[\theta])^4\,\surd\big(4\,dG^2\,(dG+tP)^2-4\,dG\,(2\,dG^3+4\,dG^2\,tP+$$
$$3\,dG\,tP^2+tP^3)\,\text{Cos}[\theta]+(2\,dG^2+2\,dG\,tP+tP^2)^2\,\text{Cos}[\theta]^2+$$
$$tP^2\,(2\,dG+tP)^2\,\text{Sin}[\theta]^2\,\text{Sin}[\phi]^2\big)\,\Big(\Big(dG+\frac{tP\,(2\,dG+tP)\,\text{Cos}[\theta]}{-2\,(dG+tP)+2\,dG\,\text{Cos}[\theta]}\Big)^4+$$
$$\big(tP^2\,(2\,dG+tP)^2\,\big(-2\,dG\,(dG+tP)+(2\,dG^2+2\,dG\,tP+tP^2)\,\text{Cos}[\theta]\big)^2$$
$$\text{Cos}[\phi]^2\,\text{Sin}[\theta]^2\big)\Big/\big(16\,(dG+tP-dG\,\text{Cos}[\theta])^4\big)-$$
$$\frac{tP^4\,(2\,dG+tP)^4\,\text{Cos}[\phi]^2\,\text{Sin}[\theta]^4\,\text{Sin}[\phi]^2}{16\,(dG+tP-dG\,\text{Cos}[\theta])^4}\Big)\Big)\Big/$$
$$\text{Abs}\big[-2\,dG\,(dG+tP)+(2\,dG^2+2\,dG\,tP+tP^2)\,\text{Cos}[\theta]\big]\Big)+$$
$$tP\,(2\,dG+tP)\,\text{Cos}[\theta]\,\Big(\Big(4\,\big(-2\,dG\,(dG+tP)+(2\,dG^2+2\,dG\,tP+tP^2)\,\text{Cos}[\theta]\big)$$
$$\big(4\,dG^2\,(dG+tP)^2-4\,dG\,(2\,dG^3+4\,dG^2\,tP+3\,dG\,tP^2+tP^3)\,\text{Cos}[\theta]+$$
$$(2\,dG^2+2\,dG\,tP+tP^2)^2\,\text{Cos}[\theta]^2+2\,tP^2\,(2\,dG+tP)^2\,\text{Cos}[\phi]^2\,\text{Sin}[\theta]^2\big)$$
$$\text{Sin}[\phi]^2\,\surd\big(4\,dG^2\,(dG+tP)^2-4\,dG\,(2\,dG^3+4\,dG^2\,tP+3\,dG\,tP^2+tP^3)\,\text{Cos}[\theta]+$$
$$(2\,dG^2+2\,dG\,tP+tP^2)^2\,\text{Cos}[\theta]^2+tP^2\,(2\,dG+tP)^2\,\text{Sin}[\theta]^2\,\text{Sin}[\phi]^2\big)\Big)\Big/$$
$$\text{Abs}\big[-2\,dG\,(dG+tP)+(2\,dG^2+2\,dG\,tP+tP^2)\,\text{Cos}[\theta]\big]+$$
$$4\,\big(-2\,dG\,(dG+tP)+(2\,dG^2+2\,dG\,tP+tP^2)\,\text{Cos}[\theta]\big)\,\text{Cos}[\phi]^2$$
$$\big(4\,dG^2\,(dG+tP)^2-4\,dG\,(2\,dG^3+4\,dG^2\,tP+3\,dG\,tP^2+tP^3)\,\text{Cos}[\theta]+$$
$$(2\,dG^2+2\,dG\,tP+tP^2)^2\,\text{Cos}[\theta]^2+tP^2\,(2\,dG+tP)^2\,\text{Sin}[\theta]^2\,\text{Sin}[\phi]^2\big)+$$
$$\Big(tP\,(2\,dG+tP)\,\text{Cos}[\theta]\,\surd\big(4\,dG^2\,(dG+tP)^2-4\,dG\,(2\,dG^3+4\,dG^2\,tP+3\,dG\,tP^2+tP^3)$$
$$\text{Cos}[\theta]+(2\,dG^2+2\,dG\,tP+tP^2)^2\,\text{Cos}[\theta]^2+tP^2\,(2\,dG+tP)^2\,\text{Sin}[\theta]^2$$
$$\text{Sin}[\phi]^2\big)\,\Big(4\,\big(-2\,dG\,(dG+tP)+(2\,dG^2+2\,dG\,tP+tP^2)\,\text{Cos}[\theta]\big)^2$$
$$\text{Cos}[2\,\phi]-tP^2\,(2\,dG+tP)^2\,\text{Sin}[\theta]^2\,\text{Sin}[2\,\phi]^2\Big)\Big)\Big/$$

$$\mathrm{Abs}\Big[-2\,dG\,(dG+tP)+\big(2\,dG^2+2\,dG\,tP+tP^2\big)\,\mathrm{Cos}[\theta]\Big]\Big)\Big)\Big)\Big/$$

$$\Big((dG+tP-dG\,\mathrm{Cos}[\theta])^3\,\big(2\,dG^2+2\,dG\,tP+tP^2-2\,dG\,(dG+tP)\,\mathrm{Cos}[\theta]\big)$$

$$\bigg(\bigg(dG+\frac{tP\,(2\,dG+tP)\,\mathrm{Cos}[\theta]}{-2\,(dG+tP)+2\,dG\,\mathrm{Cos}[\theta]}\bigg)^2+$$

$$\frac{tP^2\,(2\,dG+tP)^2\,\mathrm{Cos}[\phi]^2\,\mathrm{Sin}[\theta]^2}{4\,(dG+tP-dG\,\mathrm{Cos}[\theta])^2}\bigg)$$

$$\sqrt{\Big(4\,dG^2\,(dG+tP)^2-4\,dG\,\big(2\,dG^3+4\,dG^2\,tP+3\,dG\,tP^2+tP^3\big)\,\mathrm{Cos}[\theta]+\big(2\,dG^2+2\,dG\,tP+tP^2\big)^2\,\mathrm{Cos}[\theta]^2+tP^2\,(2\,dG+tP)^2\,\mathrm{Sin}[\theta]^2\,\mathrm{Sin}[\phi]^2\Big)}$$

$$\bigg(\bigg(dG+\frac{tP\,(2\,dG+tP)\,\mathrm{Cos}[\theta]}{-2\,(dG+tP)+2\,dG\,\mathrm{Cos}[\theta]}\bigg)^2+\frac{tP^2\,(2\,dG+tP)^2\,\mathrm{Sin}[\theta]^2\,\mathrm{Sin}[\phi]^2}{4\,(dG+tP-dG\,\mathrm{Cos}[\theta])^2}\bigg)\bigg)\bigg)$$

In[ ]:=

$$\mathbf{Z3dcond}[\theta\_,\ dG\_,\ tP\_,\ hM\_,\ \phi\_]:=$$

$$-\frac{hM*dG}{2}\bigg(\frac{1}{32}\,\mathrm{Sin}[\theta]^2\,\bigg(-\frac{32\,\mathrm{Cos}[2\,\phi]}{dG}+\Big(\mathrm{Abs}\Big[-2\,dG\,(dG+tP)+\big(2\,dG^2+2\,dG\,tP+tP^2\big)\,\mathrm{Cos}[\theta]\Big]$$

$$\bigg(16\,(dG+tP-dG\,\mathrm{Cos}[\theta])^2\,\mathrm{Cos}[\phi]^2\,\bigg(\bigg(dG+\frac{tP\,(2\,dG+tP)\,\mathrm{Cos}[\theta]}{-2\,(dG+tP)+2\,dG\,\mathrm{Cos}[\theta]}\bigg)^2+$$

$$\frac{tP^2\,(2\,dG+tP)^2\,\mathrm{Sin}[\theta]^2\,\mathrm{Sin}[\phi]^2}{4\,(dG+tP-dG\,\mathrm{Cos}[\theta])^2}\bigg)\,\Big(tP\,(2\,dG+tP)\,\mathrm{Cos}[\theta]$$

$$\Big(-2\,dG\,(dG+tP)+\big(2\,dG^2+2\,dG\,tP+tP^2\big)\,\mathrm{Cos}[\theta]\Big)+tP^2\,(2\,dG+tP)^2$$

$$\mathrm{Sin}[\theta]^2\,\mathrm{Sin}[\phi]^2+\Big(\Big(-2\,dG\,(dG+tP)+\big(2\,dG^2+2\,dG\,tP+tP^2\big)\,\mathrm{Cos}[\theta]\Big)^2$$

$$\sqrt{\Big(4\,dG^2\,(dG+tP)^2-4\,dG\,\big(2\,dG^3+4\,dG^2\,tP+3\,dG\,tP^2+tP^3\big)\,\mathrm{Cos}[\theta]+\big(2\,dG^2+2\,dG\,tP+tP^2\big)^2\,\mathrm{Cos}[\theta]^2+tP^2\,(2\,dG+tP)^2\,\mathrm{Sin}[\theta]^2\,\mathrm{Sin}[\phi]^2\Big)}\Big)\Big/$$

$$\mathrm{Abs}\Big[-2\,dG\,(dG+tP)+\big(2\,dG^2+2\,dG\,tP+tP^2\big)\,\mathrm{Cos}[\theta]\Big]\Big)+4\,\mathrm{Sin}[\phi]^2$$

$$\bigg(\Big(tP\,(2\,dG+tP)\,\mathrm{Cos}[\theta]\,\Big(-2\,dG\,(dG+tP)+\big(2\,dG^2+2\,dG\,tP+tP^2\big)\,\mathrm{Cos}[\theta]\Big)$$

$$\Big(4\,dG^2\,(dG+tP)^2-4\,dG\,\big(2\,dG^3+4\,dG^2\,tP+3\,dG\,tP^2+tP^3\big)\,\mathrm{Cos}[\theta]+$$

$$\big(2\,dG^2+2\,dG\,tP+tP^2\big)^2\,\mathrm{Cos}[\theta]^2+2\,tP^2\,(2\,dG+tP)^2\,\mathrm{Cos}[\phi]^2\,\mathrm{Sin}[\theta]^2\Big)$$

$$\sqrt{\Big(4\,dG^2\,(dG+tP)^2-4\,dG\,\big(2\,dG^3+4\,dG^2\,tP+3\,dG\,tP^2+tP^3\big)\,\mathrm{Cos}[\theta]+\big(2\,dG^2+2\,dG\,tP+tP^2\big)^2\,\mathrm{Cos}[\theta]^2+tP^2\,(2\,dG+tP)^2\,\mathrm{Sin}[\theta]^2\,\mathrm{Sin}[\phi]^2\Big)}\Big)\Big/$$

$$\mathrm{Abs}\Big[-2\,dG\,(dG+tP)+\big(2\,dG^2+2\,dG\,tP+tP^2\big)\,\mathrm{Cos}[\theta]\Big]+tP^2\,(2\,dG+tP)^2$$

$$\mathrm{Cos}[\phi]^2\,\mathrm{Sin}[\theta]^2\,\Big(4\,dG^2\,(dG+tP)^2-4\,dG\,\big(2\,dG^3+4\,dG^2\,tP+3\,dG\,tP^2+tP^3\big)\,\mathrm{Cos}[$$

$$\theta]+\big(2\,dG^2+2\,dG\,tP+tP^2\big)^2\,\mathrm{Cos}[\theta]^2+tP^2\,(2\,dG+tP)^2\,\mathrm{Sin}[\theta]^2\,\mathrm{Sin}[\phi]^2\Big)-$$

$$\Big(16\ (dG + tP - dG\, Cos[\theta])^4 \sqrt{\Big(4\, dG^2\ (dG + tP)^2 - 4\, dG\ \big(2\, dG^3 + 4\, dG^2\, tP +}$$

$$3\, dG\, tP^2 + tP^3\big)\ Cos[\theta] + \big(2\, dG^2 + 2\, dG\, tP + tP^2\big)^2\ Cos[\theta]^2 + tP^2$$

$$(2\, dG + tP)^2\ Sin[\theta]^2\ Sin[\phi]^2\Big)\ \left(\left(dG + \frac{tP\ (2\, dG + tP)\ Cos[\theta]}{-2\ (dG + tP) + 2\, dG\, Cos[\theta]}\right)^4 +\right.$$

$$\Big(tP^2\ (2\, dG + tP)^2\ \big(-2\, dG\ (dG + tP) + \big(2\, dG^2 + 2\, dG\, tP + tP^2\big)\ Cos[\theta]\big)^2$$

$$Cos[\phi]^2\ Sin[\theta]^2\Big) \Big/ \Big(16\ (dG + tP - dG\, Cos[\theta])^4\Big) -$$

$$\left.\frac{tP^4\ (2\, dG + tP)^4\ Cos[\phi]^2\ Sin[\theta]^4\ Sin[\phi]^2}{16\ (dG + tP - dG\, Cos[\theta])^4}\right)\Bigg) \Big/\ Abs\Big[$$

$$-2\, dG\ (dG + tP) + \big(2\, dG^2 + 2\, dG\, tP + tP^2\big)\ Cos[\theta]\Big]\Bigg) + tP\ (2\, dG + tP)\ Cos[\theta]$$

$$\Big(\Big(4\ \big(-2\, dG\ (dG + tP) + \big(2\, dG^2 + 2\, dG\, tP + tP^2\big)\ Cos[\theta]\big)\ \Big(4\, dG^2\ (dG + tP)^2 -$$

$$4\, dG\ \big(2\, dG^3 + 4\, dG^2\, tP + 3\, dG\, tP^2 + tP^3\big)\ Cos[\theta] + \big(2\, dG^2 + 2\, dG\, tP + tP^2\big)^2$$

$$Cos[\theta]^2 + 2\, tP^2\ (2\, dG + tP)^2\ Cos[\phi]^2\ Sin[\theta]^2\Big)\ Sin[\phi]^2$$

$$\sqrt{\Big(4\, dG^2\ (dG + tP)^2 - 4\, dG\ \big(2\, dG^3 + 4\, dG^2\, tP + 3\, dG\, tP^2 + tP^3\big)\ Cos[\theta] +}$$

$$\big(2\, dG^2 + 2\, dG\, tP + tP^2\big)^2\ Cos[\theta]^2 + tP^2\ (2\, dG + tP)^2\ Sin[\theta]^2\ Sin[\phi]^2\Big)\Big) \Big/$$

$$Abs\Big[-2\, dG\ (dG + tP) + \big(2\, dG^2 + 2\, dG\, tP + tP^2\big)\ Cos[\theta]\Big] + 4$$

$$\big(-2\, dG\ (dG + tP) + \big(2\, dG^2 + 2\, dG\, tP + tP^2\big)\ Cos[\theta]\big)\ Cos[\phi]^2\ \Big(4\, dG^2\ (dG + tP)^2 -$$

$$4\, dG\ \big(2\, dG^3 + 4\, dG^2\, tP + 3\, dG\, tP^2 + tP^3\big)\ Cos[\theta] + \big(2\, dG^2 + 2\, dG\, tP + tP^2\big)^2$$

$$Cos[\theta]^2 + tP^2\ (2\, dG + tP)^2\ Sin[\theta]^2\ Sin[\phi]^2\Big) + \Big(tP\ (2\, dG + tP)\ Cos[\theta]$$

$$\sqrt{\Big(4\, dG^2\ (dG + tP)^2 - 4\, dG\ \big(2\, dG^3 + 4\, dG^2\, tP + 3\, dG\, tP^2 + tP^3\big)\ Cos[\theta] +}$$

$$\big(2\, dG^2 + 2\, dG\, tP + tP^2\big)^2\ Cos[\theta]^2 + tP^2\ (2\, dG + tP)^2\ Sin[\theta]^2\ Sin[\phi]^2\Big)$$

$$\Big(4\ \big(-2\, dG\ (dG + tP) + \big(2\, dG^2 + 2\, dG\, tP + tP^2\big)\ Cos[\theta]\big)^2\ Cos[2\, \phi] -$$

$$tP^2\ (2\, dG + tP)^2\ Sin[\theta]^2\ Sin[2\, \phi]^2\Big)\Big) \Big/\ Abs\Big[$$

$$-2\, dG\ (dG + tP) + \big(2\, dG^2 + 2\, dG\, tP + tP^2\big)\ Cos[\theta]\Big]\Big)\Big)\Big) \Big/$$

$$\Big((dG + tP - dG\, Cos[\theta])^3\ \big(2\, dG^2 + 2\, dG\, tP + tP^2 - 2\, dG\ (dG + tP)\ Cos[\theta]\big)$$

$$\left(\left(dG + \frac{tP\ (2\, dG + tP)\ Cos[\theta]}{-2\ (dG + tP) + 2\, dG\, Cos[\theta]}\right)^2 +\right.$$

$$\left.\frac{tP^2\ (2\, dG + tP)^2\ Cos[\phi]^2\ Sin[\theta]^2}{4\ (dG + tP - dG\, Cos[\theta])^2}\right)$$

$$\sqrt{\Big(4\, dG^2\ (dG + tP)^2 - 4\, dG\ \big(2\, dG^3 + 4\, dG^2\, tP + 3\, dG\, tP^2 + tP^3\big)\ Cos[\theta] +}$$

$$\big(2\, dG^2 + 2\, dG\, tP + tP^2\big)^2\ Cos[\theta]^2 + tP^2\ (2\, dG + tP)^2\ Sin[\theta]^2\ Sin[\phi]^2\Big)$$

$$\left(\left(\left(\text{dG}+\frac{\text{tP}\ (2\ \text{dG}+\text{tP})\ \text{Cos}[\theta]}{-2\ (\text{dG}+\text{tP})+2\ \text{dG}\ \text{Cos}[\theta]}\right)^2+\frac{\text{tP}^2\ (2\ \text{dG}+\text{tP})^2\ \text{Sin}[\theta]^2\ \text{Sin}[\phi]^2}{4\ (\text{dG}+\text{tP}-\text{dG}\ \text{Cos}[\theta])^2}\right)\right)\right)\right)$$

In[∘]:=

$$\text{Simplify}\Big[\frac{1}{32}\ \text{Sin}[\theta]^2\ \Big(-\frac{32\ \text{Cos}[2\ \phi]}{\text{dG}}+$$

$$\Big(\text{Abs}\left[-2\ \text{dG}\ (\text{dG}+\text{tP})+\left(2\ \text{dG}^2+2\ \text{dG}\ \text{tP}+\text{tP}^2\right)\ \text{Cos}[\theta]\right]\ \Big(16\ (\text{dG}+\text{tP}-\text{dG}\ \text{Cos}[\theta])^2\ \text{Cos}[\phi]^2$$

$$\left(\left(\text{dG}+\frac{\text{tP}\ (2\ \text{dG}+\text{tP})\ \text{Cos}[\theta]}{-2\ (\text{dG}+\text{tP})+2\ \text{dG}\ \text{Cos}[\theta]}\right)^2+\frac{\text{tP}^2\ (2\ \text{dG}+\text{tP})^2\ \text{Sin}[\theta]^2\ \text{Sin}[\phi]^2}{4\ (\text{dG}+\text{tP}-\text{dG}\ \text{Cos}[\theta])^2}\right)$$

$$\Big(\text{tP}\ (2\ \text{dG}+\text{tP})\ \text{Cos}[\theta]\ \left(-2\ \text{dG}\ (\text{dG}+\text{tP})+\left(2\ \text{dG}^2+2\ \text{dG}\ \text{tP}+\text{tP}^2\right)\ \text{Cos}[\theta]\right)+\text{tP}^2\ (2\ \text{dG}+$$

$$\text{tP})^2\ \text{Sin}[\theta]^2\ \text{Sin}[\phi]^2+\Big(\left(-2\ \text{dG}\ (\text{dG}+\text{tP})+\left(2\ \text{dG}^2+2\ \text{dG}\ \text{tP}+\text{tP}^2\right)\ \text{Cos}[\theta]\right)^2$$

$$\sqrt{\Big(4\ \text{dG}^2\ (\text{dG}+\text{tP})^2-4\ \text{dG}\ \left(2\ \text{dG}^3+4\ \text{dG}^2\ \text{tP}+3\ \text{dG}\ \text{tP}^2+\text{tP}^3\right)\ \text{Cos}[\theta]+}$$

$$\left.\left(2\ \text{dG}^2+2\ \text{dG}\ \text{tP}+\text{tP}^2\right)^2\ \text{Cos}[\theta]^2+\text{tP}^2\ (2\ \text{dG}+\text{tP})^2\ \text{Sin}[\theta]^2\ \text{Sin}[\phi]^2\Big)\Big)\right/$$

$$\text{Abs}\left[-2\ \text{dG}\ (\text{dG}+\text{tP})+\left(2\ \text{dG}^2+2\ \text{dG}\ \text{tP}+\text{tP}^2\right)\ \text{Cos}[\theta]\right]\Big)+4\ \text{Sin}[\phi]^2$$

$$\Big(\Big(\text{tP}\ (2\ \text{dG}+\text{tP})\ \text{Cos}[\theta]\ \left(-2\ \text{dG}\ (\text{dG}+\text{tP})+\left(2\ \text{dG}^2+2\ \text{dG}\ \text{tP}+\text{tP}^2\right)\ \text{Cos}[\theta]\right)$$

$$\Big(4\ \text{dG}^2\ (\text{dG}+\text{tP})^2-4\ \text{dG}\ \left(2\ \text{dG}^3+4\ \text{dG}^2\ \text{tP}+3\ \text{dG}\ \text{tP}^2+\text{tP}^3\right)\ \text{Cos}[\theta]+$$

$$\left(2\ \text{dG}^2+2\ \text{dG}\ \text{tP}+\text{tP}^2\right)^2\ \text{Cos}[\theta]^2+2\ \text{tP}^2\ (2\ \text{dG}+\text{tP})^2\ \text{Cos}[\phi]^2\ \text{Sin}[\theta]^2\Big)$$

$$\sqrt{\Big(4\ \text{dG}^2\ (\text{dG}+\text{tP})^2-4\ \text{dG}\ \left(2\ \text{dG}^3+4\ \text{dG}^2\ \text{tP}+3\ \text{dG}\ \text{tP}^2+\text{tP}^3\right)\ \text{Cos}[\theta]+}$$

$$\left.\left(2\ \text{dG}^2+2\ \text{dG}\ \text{tP}+\text{tP}^2\right)^2\ \text{Cos}[\theta]^2+\text{tP}^2\ (2\ \text{dG}+\text{tP})^2\ \text{Sin}[\theta]^2\ \text{Sin}[\phi]^2\Big)\Big)\right/$$

$$\text{Abs}\left[-2\ \text{dG}\ (\text{dG}+\text{tP})+\left(2\ \text{dG}^2+2\ \text{dG}\ \text{tP}+\text{tP}^2\right)\ \text{Cos}[\theta]\right]+\text{tP}^2\ (2\ \text{dG}+\text{tP})^2$$

$$\text{Cos}[\phi]^2\ \text{Sin}[\theta]^2\ \Big(4\ \text{dG}^2\ (\text{dG}+\text{tP})^2-4\ \text{dG}\ \left(2\ \text{dG}^3+4\ \text{dG}^2\ \text{tP}+3\ \text{dG}\ \text{tP}^2+\text{tP}^3\right)\ \text{Cos}[$$

$$\theta]+\left(2\ \text{dG}^2+2\ \text{dG}\ \text{tP}+\text{tP}^2\right)^2\ \text{Cos}[\theta]^2+\text{tP}^2\ (2\ \text{dG}+\text{tP})^2\ \text{Sin}[\theta]^2\ \text{Sin}[\phi]^2\Big)-$$

$$\Big(16\ (\text{dG}+\text{tP}-\text{dG}\ \text{Cos}[\theta])^4\ \sqrt{\Big(4\ \text{dG}^2\ (\text{dG}+\text{tP})^2-4\ \text{dG}\ \Big(2\ \text{dG}^3+4\ \text{dG}^2\ \text{tP}+}$$

$$3\ \text{dG}\ \text{tP}^2+\text{tP}^3\Big)\ \text{Cos}[\theta]+\left(2\ \text{dG}^2+2\ \text{dG}\ \text{tP}+\text{tP}^2\right)^2\ \text{Cos}[\theta]^2+$$

$$\text{tP}^2\ (2\ \text{dG}+\text{tP})^2\ \text{Sin}[\theta]^2\ \text{Sin}[\phi]^2\Big)\ \left(\left(\text{dG}+\frac{\text{tP}\ (2\ \text{dG}+\text{tP})\ \text{Cos}[\theta]}{-2\ (\text{dG}+\text{tP})+2\ \text{dG}\ \text{Cos}[\theta]}\right)^4+\right.$$

$$\Big(\text{tP}^2\ (2\ \text{dG}+\text{tP})^2\ \left(-2\ \text{dG}\ (\text{dG}+\text{tP})+\left(2\ \text{dG}^2+2\ \text{dG}\ \text{tP}+\text{tP}^2\right)\ \text{Cos}[\theta]\right)^2$$

$$\left.\text{Cos}[\phi]^2\ \text{Sin}[\theta]^2\Big)\right/\left(16\ (\text{dG}+\text{tP}-\text{dG}\ \text{Cos}[\theta])^4\right)-$$

$$\left.\left.\frac{\text{tP}^4\ (2\ \text{dG}+\text{tP})^4\ \text{Cos}[\phi]^2\ \text{Sin}[\theta]^4\ \text{Sin}[\phi]^2}{16\ (\text{dG}+\text{tP}-\text{dG}\ \text{Cos}[\theta])^4}\right)\Big)\right/$$

$$\text{Abs}\left[-2\ \text{dG}\ (\text{dG}+\text{tP})+\left(2\ \text{dG}^2+2\ \text{dG}\ \text{tP}+\text{tP}^2\right)\ \text{Cos}[\theta]\right]\Big)+$$

$$\text{tP}\ (2\ \text{dG}+\text{tP})\ \text{Cos}[\theta]\ \Big(\Big(4\ \left(-2\ \text{dG}\ (\text{dG}+\text{tP})+\left(2\ \text{dG}^2+2\ \text{dG}\ \text{tP}+\text{tP}^2\right)\ \text{Cos}[\theta]\right)$$

```
(4 dG^2 (dG + tP)^2 - 4 dG (2 dG^3 + 4 dG^2 tP + 3 dG tP^2 + tP^3) Cos[θ] +
    (2 dG^2 + 2 dG tP + tP^2)^2 Cos[θ]^2 + 2 tP^2 (2 dG + tP)^2 Cos[ϕ]^2 Sin[θ]^2)
  Sin[ϕ]^2 √(4 dG^2 (dG + tP)^2 - 4 dG (2 dG^3 + 4 dG^2 tP + 3 dG tP^2 + tP^3) Cos[θ] +
      (2 dG^2 + 2 dG tP + tP^2)^2 Cos[θ]^2 + tP^2 (2 dG + tP)^2 Sin[θ]^2 Sin[ϕ]^2)) /
 Abs[-2 dG (dG + tP) + (2 dG^2 + 2 dG tP + tP^2) Cos[θ]] +
 4 (-2 dG (dG + tP) + (2 dG^2 + 2 dG tP + tP^2) Cos[θ]) Cos[ϕ]^2
 (4 dG^2 (dG + tP)^2 - 4 dG (2 dG^3 + 4 dG^2 tP + 3 dG tP^2 + tP^3) Cos[θ] +
   (2 dG^2 + 2 dG tP + tP^2)^2 Cos[θ]^2 + tP^2 (2 dG + tP)^2 Sin[θ]^2 Sin[ϕ]^2) +
 (tP (2 dG + tP) Cos[θ] √(4 dG^2 (dG + tP)^2 - 4 dG (2 dG^3 + 4 dG^2 tP + 3 dG tP^2 + tP^3)
      Cos[θ] + (2 dG^2 + 2 dG tP + tP^2)^2 Cos[θ]^2 + tP^2 (2 dG + tP)^2 Sin[θ]^2
      Sin[ϕ]^2) (4 (-2 dG (dG + tP) + (2 dG^2 + 2 dG tP + tP^2) Cos[θ])^2
     Cos[2 ϕ] - tP^2 (2 dG + tP)^2 Sin[θ]^2 Sin[2 ϕ]^2)) /
 Abs[-2 dG (dG + tP) + (2 dG^2 + 2 dG tP + tP^2) Cos[θ]])) /

((dG + tP - dG Cos[θ])^3 (2 dG^2 + 2 dG tP + tP^2 - 2 dG (dG + tP) Cos[θ])
  (((dG + (tP (2 dG + tP) Cos[θ])/(-2 (dG + tP) + 2 dG Cos[θ]))^2 +
    (tP^2 (2 dG + tP)^2 Cos[ϕ]^2 Sin[θ]^2)/(4 (dG + tP - dG Cos[θ])^2))
  √(4 dG^2 (dG + tP)^2 - 4 dG (2 dG^3 + 4 dG^2 tP + 3 dG tP^2 + tP^3) Cos[θ] +
     (2 dG^2 + 2 dG tP + tP^2)^2 Cos[θ]^2 + tP^2 (2 dG + tP)^2 Sin[θ]^2 Sin[ϕ]^2)
  (((dG + (tP (2 dG + tP) Cos[θ])/(-2 (dG + tP) + 2 dG Cos[θ]))^2 +
    (tP^2 (2 dG + tP)^2 Sin[θ]^2 Sin[ϕ]^2)/(4 (dG + tP - dG Cos[θ])^2)))), Assumptions → tP > dG]

In[ ]:= Simplify[1/32 Sin[θ]^2 (-(32 Cos[2 ϕ])/dG +
    (Abs[-2 dG (dG + tP) + (2 dG^2 + 2 dG tP + tP^2) Cos[θ]] (16 (dG + tP - dG Cos[θ])^2 Cos[ϕ]^2
        (((dG + (tP (2 dG + tP) Cos[θ])/(-2 (dG + tP) + 2 dG Cos[θ]))^2 + (tP^2 (2 dG + tP)^2 Sin[θ]^2 Sin[ϕ]^2)/(4 (dG + tP - dG Cos[θ])^2))
        (tP (2 dG + tP) Cos[θ] (-2 dG (dG + tP) + (2 dG^2 + 2 dG tP + tP^2) Cos[θ]) + tP^2 (2 dG +
             tP)^2 Sin[θ]^2 Sin[ϕ]^2 + ((-2 dG (dG + tP) + (2 dG^2 + 2 dG tP + tP^2) Cos[θ])^2
            √(4 dG^2 (dG + tP)^2 - 4 dG (2 dG^3 + 4 dG^2 tP + 3 dG tP^2 + tP^3) Cos[θ] +
```

$$
\begin{aligned}
&\left(2\,dG^2+2\,dG\,tP+tP^2\right)^2 Cos[\theta]^2+tP^2\,(2\,dG+tP)^2\,Sin[\theta]^2\,Sin[\phi]^2\Big)\Big)\Big/\\
&Abs\left[-2\,dG\,(dG+tP)+\left(2\,dG^2+2\,dG\,tP+tP^2\right)Cos[\theta]\right]\Big)+4\,Sin[\phi]^2\\
&\Big(\Big(tP\,(2\,dG+tP)\,Cos[\theta]\,\left(-2\,dG\,(dG+tP)+\left(2\,dG^2+2\,dG\,tP+tP^2\right)Cos[\theta]\right)\\
&\Big(4\,dG^2\,(dG+tP)^2-4\,dG\left(2\,dG^3+4\,dG^2\,tP+3\,dG\,tP^2+tP^3\right)Cos[\theta]+\\
&\left(2\,dG^2+2\,dG\,tP+tP^2\right)^2 Cos[\theta]^2+2\,tP^2\,(2\,dG+tP)^2\,Cos[\phi]^2\,Sin[\theta]^2\Big)\\
&\sqrt{\Big(4\,dG^2\,(dG+tP)^2-4\,dG\left(2\,dG^3+4\,dG^2\,tP+3\,dG\,tP^2+tP^3\right)Cos[\theta]+}\\
&\left(2\,dG^2+2\,dG\,tP+tP^2\right)^2 Cos[\theta]^2+tP^2\,(2\,dG+tP)^2\,Sin[\theta]^2\,Sin[\phi]^2\Big)\Big)\Big/\\
&Abs\left[-2\,dG\,(dG+tP)+\left(2\,dG^2+2\,dG\,tP+tP^2\right)Cos[\theta]\right]+tP^2\,(2\,dG+tP)^2\\
&Cos[\phi]^2\,Sin[\theta]^2\,\Big(4\,dG^2\,(dG+tP)^2-4\,dG\left(2\,dG^3+4\,dG^2\,tP+3\,dG\,tP^2+tP^3\right)Cos[\\
&\theta]+\left(2\,dG^2+2\,dG\,tP+tP^2\right)^2 Cos[\theta]^2+tP^2\,(2\,dG+tP)^2\,Sin[\theta]^2\,Sin[\phi]^2\Big)-\\
&\Big(16\,(dG+tP-dG\,Cos[\theta])^4\sqrt{\Big(4\,dG^2\,(dG+tP)^2-4\,dG\,\Big(2\,dG^3+4\,dG^2\,tP+}\\
&3\,dG\,tP^2+tP^3\Big)Cos[\theta]+\left(2\,dG^2+2\,dG\,tP+tP^2\right)^2 Cos[\theta]^2+\\
&tP^2\,(2\,dG+tP)^2\,Sin[\theta]^2\,Sin[\phi]^2\Big)\,\left(\left(dG+\frac{tP\,(2\,dG+tP)\,Cos[\theta]}{-2\,(dG+tP)+2\,dG\,Cos[\theta]}\right)^4+\right.\\
&\Big(tP^2\,(2\,dG+tP)^2\left(-2\,dG\,(dG+tP)+\left(2\,dG^2+2\,dG\,tP+tP^2\right)Cos[\theta]\right)^2\\
&Cos[\phi]^2\,Sin[\theta]^2\Big)\Big/\left(16\,(dG+tP-dG\,Cos[\theta])^4\right)-\\
&\left.\frac{tP^4\,(2\,dG+tP)^4\,Cos[\phi]^2\,Sin[\theta]^4\,Sin[\phi]^2}{16\,(dG+tP-dG\,Cos[\theta])^4}\right)\Big)\Big/\\
&Abs\left[-2\,dG\,(dG+tP)+\left(2\,dG^2+2\,dG\,tP+tP^2\right)Cos[\theta]\right]\Big)+\\
&tP\,(2\,dG+tP)\,Cos[\theta]\,\Big(\Big(4\left(-2\,dG\,(dG+tP)+\left(2\,dG^2+2\,dG\,tP+tP^2\right)Cos[\theta]\right)\\
&\Big(4\,dG^2\,(dG+tP)^2-4\,dG\left(2\,dG^3+4\,dG^2\,tP+3\,dG\,tP^2+tP^3\right)Cos[\theta]+\\
&\left(2\,dG^2+2\,dG\,tP+tP^2\right)^2 Cos[\theta]^2+2\,tP^2\,(2\,dG+tP)^2\,Cos[\phi]^2\,Sin[\theta]^2\Big)\\
&Sin[\phi]^2\sqrt{\Big(4\,dG^2\,(dG+tP)^2-4\,dG\left(2\,dG^3+4\,dG^2\,tP+3\,dG\,tP^2+tP^3\right)Cos[\theta]+}\\
&\left(2\,dG^2+2\,dG\,tP+tP^2\right)^2 Cos[\theta]^2+tP^2\,(2\,dG+tP)^2\,Sin[\theta]^2\,Sin[\phi]^2\Big)\Big)\Big/\\
&Abs\left[-2\,dG\,(dG+tP)+\left(2\,dG^2+2\,dG\,tP+tP^2\right)Cos[\theta]\right]+\\
&4\left(-2\,dG\,(dG+tP)+\left(2\,dG^2+2\,dG\,tP+tP^2\right)Cos[\theta]\right)Cos[\phi]^2\\
&\Big(4\,dG^2\,(dG+tP)^2-4\,dG\left(2\,dG^3+4\,dG^2\,tP+3\,dG\,tP^2+tP^3\right)Cos[\theta]+\\
&\left(2\,dG^2+2\,dG\,tP+tP^2\right)^2 Cos[\theta]^2+tP^2\,(2\,dG+tP)^2\,Sin[\theta]^2\,Sin[\phi]^2\Big)+\\
&\Big(tP\,(2\,dG+tP)\,Cos[\theta]\sqrt{\Big(4\,dG^2\,(dG+tP)^2-4\,dG\left(2\,dG^3+4\,dG^2\,tP+3\,dG\,tP^2+tP^3\right)}\\
&Cos[\theta]+\left(2\,dG^2+2\,dG\,tP+tP^2\right)^2 Cos[\theta]^2+tP^2\,(2\,dG+tP)^2\,Sin[\theta]^2\\
&Sin[\phi]^2\Big)\,\Big(4\left(-2\,dG\,(dG+tP)+\left(2\,dG^2+2\,dG\,tP+tP^2\right)Cos[\theta]\right)^2
\end{aligned}
$$

```
Cos[2 φ] - tP^2 (2 dG + tP)^2 Sin[θ]^2 Sin[2 φ]^2)) /
      Abs[-2 dG (dG + tP) + (2 dG^2 + 2 dG tP + tP^2) Cos[θ]]))) /
  ((dG + tP - dG Cos[θ])^3 (2 dG^2 + 2 dG tP + tP^2 - 2 dG (dG + tP) Cos[θ])
    ((dG + (tP (2 dG + tP) Cos[θ]) / (-2 (dG + tP) + 2 dG Cos[θ]))^2 +
      (tP^2 (2 dG + tP)^2 Cos[ϕ]^2 Sin[θ]^2) / (4 (dG + tP - dG Cos[θ])^2))
    √(4 dG^2 (dG + tP)^2 - 4 dG (2 dG^3 + 4 dG^2 tP + 3 dG tP^2 + tP^3) Cos[θ] +
        (2 dG^2 + 2 dG tP + tP^2)^2 Cos[θ]^2 + tP^2 (2 dG + tP)^2 Sin[θ]^2 Sin[ϕ]^2)
    ((dG + (tP (2 dG + tP) Cos[θ]) / (-2 (dG + tP) + 2 dG Cos[θ]))^2 + (tP^2 (2 dG + tP)^2 Sin[θ]^2 Sin[ϕ]^2) / (4 (dG + tP - dG Cos[θ])^2)))),
 Assumptions → {tP > dG, dG /
     tP →
    0}]
```

Out[ ]=

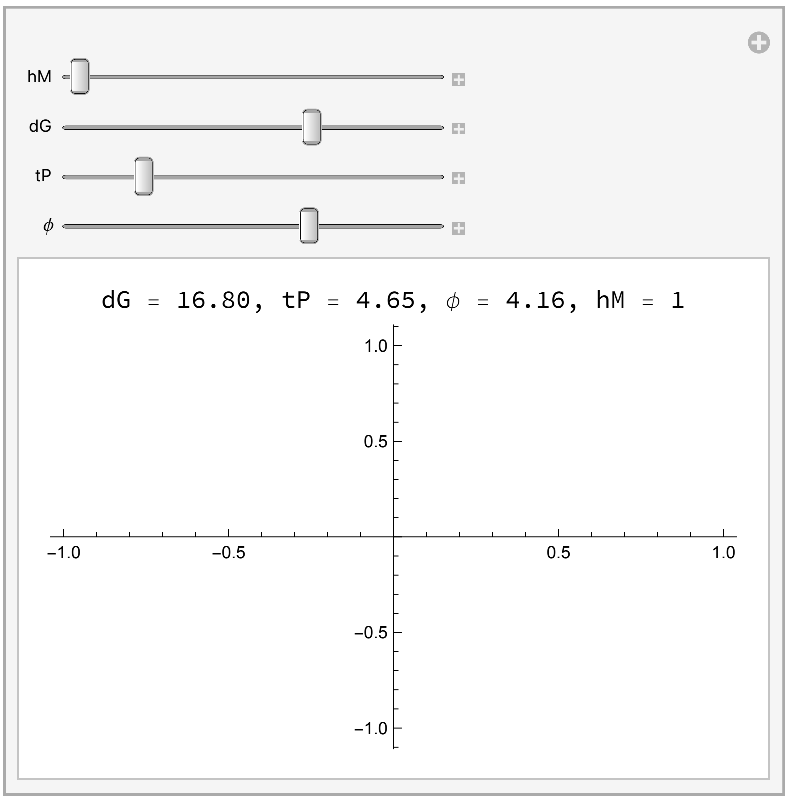
```
1/32 Sin[θ]^2 (-(32 Cos[2 ϕ]) / dG +
   (Abs[-2 dG (dG + tP) + (2 dG^2 + 2 dG tP + tP^2) Cos[θ]] (16 (dG + tP - dG Cos[θ])^2 Cos[ϕ]^2
        ((dG + (tP (2 dG + tP) Cos[θ]) / (-2 (dG + tP) + 2 dG Cos[θ]))^2 + (tP^2 (2 dG + tP)^2 Sin[θ]^2 Sin[ϕ]^2) / (4 (dG + tP - dG Cos[θ])^2))
        (tP (2 dG + tP) Cos[θ] (-2 dG (dG + tP) + (2 dG^2 + 2 dG tP + tP^2) Cos[θ]) + tP^2 (2 dG +
              tP)^2 Sin[θ]^2 Sin[ϕ]^2 + ((-2 dG (dG + tP) + (2 dG^2 + 2 dG tP + tP^2) Cos[θ])^2
              √(4 dG^2 (dG + tP)^2 - 4 dG (2 dG^3 + 4 dG^2 tP + 3 dG tP^2 + tP^3) Cos[θ] +
                 (2 dG^2 + 2 dG tP + tP^2)^2 Cos[θ]^2 + tP^2 (2 dG + tP)^2 Sin[θ]^2 Sin[ϕ]^2)) /
            Abs[-2 dG (dG + tP) + (2 dG^2 + 2 dG tP + tP^2) Cos[θ]]) +
       4 Sin[ϕ]^2 ((tP (2 dG + tP) Cos[θ] (-2 dG (dG + tP) + (2 dG^2 + 2 dG tP + tP^2) Cos[θ])
            (4 dG^2 (dG + tP)^2 - 4 dG (2 dG^3 + 4 dG^2 tP + 3 dG tP^2 + tP^3) Cos[θ] +
              (2 dG^2 + 2 dG tP + tP^2)^2 Cos[θ]^2 + 2 tP^2 (2 dG + tP)^2 Cos[ϕ]^2 Sin[θ]^2)
            √(4 dG^2 (dG + tP)^2 - 4 dG (2 dG^3 + 4 dG^2 tP + 3 dG tP^2 + tP^3) Cos[θ] +
               (2 dG^2 + 2 dG tP + tP^2)^2 Cos[θ]^2 + tP^2 (2 dG + tP)^2 Sin[θ]^2 Sin[ϕ]^2)) /
          Abs[-2 dG (dG + tP) + (2 dG^2 + 2 dG tP + tP^2) Cos[θ]] + tP^2 (2 dG + tP)^2 Cos[ϕ]^2
```

$$\mathrm{Sin}[\theta]^2\left(4\,dG^2\,(dG+tP)^2-4\,dG\left(2\,dG^3+4\,dG^2\,tP+3\,dG\,tP^2+tP^3\right)\mathrm{Cos}[\theta]+\right.$$
$$\left.\left(2\,dG^2+2\,dG\,tP+tP^2\right)^2\mathrm{Cos}[\theta]^2+tP^2\,(2\,dG+tP)^2\,\mathrm{Sin}[\theta]^2\,\mathrm{Sin}[\phi]^2\right)-$$
$$\left(16\,(dG+tP-dG\,\mathrm{Cos}[\theta])^4\sqrt{\left(4\,dG^2\,(dG+tP)^2-4\,dG\left(2\,dG^3+4\,dG^2\,tP+\right.\right.}\right.$$
$$3\,dG\,tP^2+tP^3\big)\,\mathrm{Cos}[\theta]+\left(2\,dG^2+2\,dG\,tP+tP^2\right)^2\mathrm{Cos}[\theta]^2+$$
$$tP^2\,(2\,dG+tP)^2\,\mathrm{Sin}[\theta]^2\,\mathrm{Sin}[\phi]^2\Big)\left(\left(dG+\frac{tP\,(2\,dG+tP)\,\mathrm{Cos}[\theta]}{-2\,(dG+tP)+2\,dG\,\mathrm{Cos}[\theta]}\right)^4+\right.$$
$$\left(tP^2\,(2\,dG+tP)^2\left(-2\,dG\,(dG+tP)+\left(2\,dG^2+2\,dG\,tP+tP^2\right)\mathrm{Cos}[\theta]\right)^2\right.$$
$$\left.\mathrm{Cos}[\phi]^2\,\mathrm{Sin}[\theta]^2\right)\Big/\left(16\,(dG+tP-dG\,\mathrm{Cos}[\theta])^4\right)-$$
$$\left.\left.\frac{tP^4\,(2\,dG+tP)^4\,\mathrm{Cos}[\phi]^2\,\mathrm{Sin}[\theta]^4\,\mathrm{Sin}[\phi]^2}{16\,(dG+tP-dG\,\mathrm{Cos}[\theta])^4}\right)\right)\Big/$$
$$\left.\mathrm{Abs}\left[-2\,dG\,(dG+tP)+\left(2\,dG^2+2\,dG\,tP+tP^2\right)\mathrm{Cos}[\theta]\right]\right)+$$
$$tP\,(2\,dG+tP)\,\mathrm{Cos}[\theta]\,\left(\left(4\left(-2\,dG\,(dG+tP)+\left(2\,dG^2+2\,dG\,tP+tP^2\right)\mathrm{Cos}[\theta]\right)\right.\right.$$
$$\left(4\,dG^2\,(dG+tP)^2-4\,dG\left(2\,dG^3+4\,dG^2\,tP+3\,dG\,tP^2+tP^3\right)\mathrm{Cos}[\theta]+\right.$$
$$\left.\left(2\,dG^2+2\,dG\,tP+tP^2\right)^2\mathrm{Cos}[\theta]^2+2\,tP^2\,(2\,dG+tP)^2\,\mathrm{Cos}[\phi]^2\,\mathrm{Sin}[\theta]^2\right)$$
$$\mathrm{Sin}[\phi]^2\sqrt{\left(4\,dG^2\,(dG+tP)^2-4\,dG\left(2\,dG^3+4\,dG^2\,tP+3\,dG\,tP^2+tP^3\right)\mathrm{Cos}[\theta]+\right.}$$
$$\left.\left.\left(2\,dG^2+2\,dG\,tP+tP^2\right)^2\mathrm{Cos}[\theta]^2+tP^2\,(2\,dG+tP)^2\,\mathrm{Sin}[\theta]^2\,\mathrm{Sin}[\phi]^2\right)\right)\Big/$$
$$\mathrm{Abs}\left[-2\,dG\,(dG+tP)+\left(2\,dG^2+2\,dG\,tP+tP^2\right)\mathrm{Cos}[\theta]\right]+$$
$$4\left(-2\,dG\,(dG+tP)+\left(2\,dG^2+2\,dG\,tP+tP^2\right)\mathrm{Cos}[\theta]\right)\mathrm{Cos}[\phi]^2$$
$$\left(4\,dG^2\,(dG+tP)^2-4\,dG\left(2\,dG^3+4\,dG^2\,tP+3\,dG\,tP^2+tP^3\right)\mathrm{Cos}[\theta]+\right.$$
$$\left.\left(2\,dG^2+2\,dG\,tP+tP^2\right)^2\mathrm{Cos}[\theta]^2+tP^2\,(2\,dG+tP)^2\,\mathrm{Sin}[\theta]^2\,\mathrm{Sin}[\phi]^2\right)+$$
$$\left(tP\,(2\,dG+tP)\,\mathrm{Cos}[\theta]\sqrt{\left(4\,dG^2\,(dG+tP)^2-4\,dG\left(2\,dG^3+4\,dG^2\,tP+3\,dG\,tP^2+tP^3\right)\right.}\right.$$
$$\mathrm{Cos}[\theta]+\left(2\,dG^2+2\,dG\,tP+tP^2\right)^2\mathrm{Cos}[\theta]^2+tP^2\,(2\,dG+tP)^2\,\mathrm{Sin}[\theta]^2$$
$$\left.\mathrm{Sin}[\phi]^2\right)\left(4\left(-2\,dG\,(dG+tP)+\left(2\,dG^2+2\,dG\,tP+tP^2\right)\mathrm{Cos}[\theta]\right)^2\right.$$
$$\left.\left.\mathrm{Cos}[2\,\phi]-tP^2\,(2\,dG+tP)^2\,\mathrm{Sin}[\theta]^2\,\mathrm{Sin}[2\,\phi]^2\right)\right)\Big/$$
$$\left.\left.\mathrm{Abs}\left[-2\,dG\,(dG+tP)+\left(2\,dG^2+2\,dG\,tP+tP^2\right)\mathrm{Cos}[\theta]\right]\right)\right)\Big/$$
$$\left((dG+tP-dG\,\mathrm{Cos}[\theta])^3\left(2\,dG^2+2\,dG\,tP+tP^2-2\,dG\,(dG+tP)\,\mathrm{Cos}[\theta]\right)\right.$$
$$\left(\left(dG+\frac{tP\,(2\,dG+tP)\,\mathrm{Cos}[\theta]}{-2\,(dG+tP)+2\,dG\,\mathrm{Cos}[\theta]}\right)^2+\right.$$
$$\left.\frac{tP^2\,(2\,dG+tP)^2\,\mathrm{Cos}[\phi]^2\,\mathrm{Sin}[\theta]^2}{4\,(dG+tP-dG\,\mathrm{Cos}[\theta])^2}\right)$$
$$\sqrt{\left(4\,dG^2\,(dG+tP)^2-4\,dG\left(2\,dG^3+4\,dG^2\,tP+3\,dG\,tP^2+tP^3\right)\mathrm{Cos}[\theta]+\right.}$$

$$\left(2\,dG^2 + 2\,dG\,tP + tP^2\right)^2 \mathrm{Cos}[\theta]^2 +$$
$$tP^2\,(2\,dG + tP)^2\,\mathrm{Sin}[\theta]^2\,\mathrm{Sin}[\phi]^2\Big)$$
$$\left(\left(dG + \frac{tP\,(2\,dG + tP)\,\mathrm{Cos}[\theta]}{-2\,(dG + tP) + 2\,dG\,\mathrm{Cos}[\theta]}\right)^2 + \frac{tP^2\,(2\,dG + tP)^2\,\mathrm{Sin}[\theta]^2\,\mathrm{Sin}[\phi]^2}{4\,(dG + tP - dG\,\mathrm{Cos}[\theta])^2}\right)\Bigg)\Bigg)$$

In[∘]:=
```
Manipulate[Labeled[Plot[Z3dcond[θ * π, dG, tP, hM, ϕ], {θ, 0, 1}, PlotRange → All],
  Row[{"dG = ", NumberForm[dG, {3, 2}], ", tP = ", NumberForm[tP, {3, 2}],
    ", ϕ = ", NumberForm[ϕ, {3, 2}], ", hM = ", NumberForm[hM, {3, 2}]}], Top],
 {hM, 1, 2}, {dG, 0.1, 25}, {tP, 0.01, 25}, {ϕ, 0, 2 π}]
```

Out[∘]=



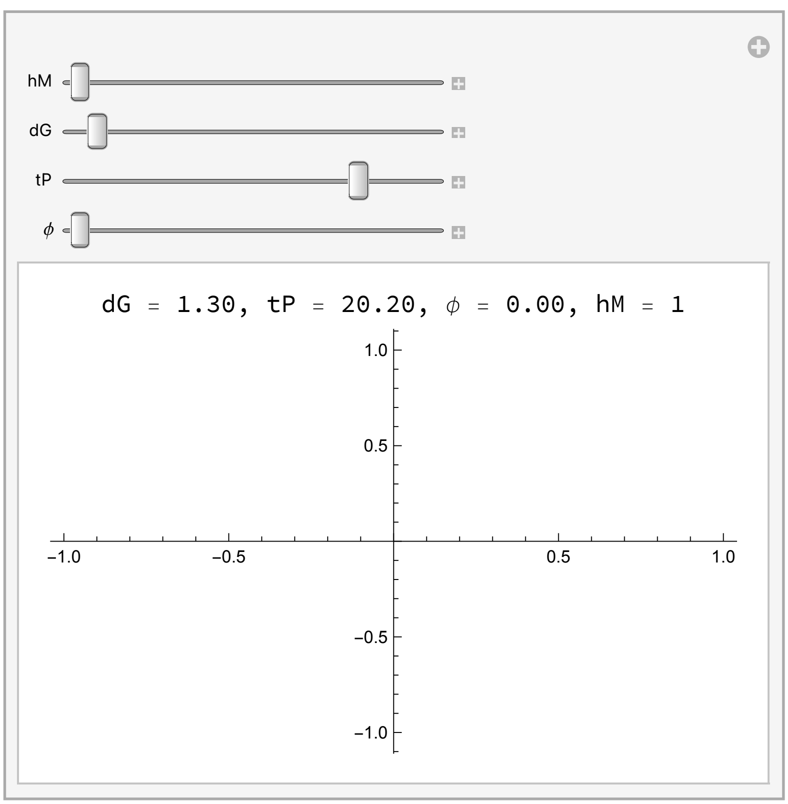
In[∘]:=

```
Z3dfull[θ_, dG_, tP_, hM_, ϕ_] :=
  If[(((θ ≤ ArcCos[dG / (tP - (tP^2 - 2 dG tP Cos[θ]) / (2 dG + 2 tP - 2 dG Cos[θ]))] || θ + ArcCos[dG / (tP - (tP^2 - 2 dG tP Cos[θ]) / (2 dG + 2 tP - 2 dG Cos[θ]))] > 2 π) &&
      dG + (tP^2 - 2 dG tP Cos[θ]) / (2 dG + 2 tP - 2 dG Cos[θ]) < tP),
   Z3dcond[θ, dG, tP, hM, ϕ], Z3d2[θ, dG, tP, hM, ϕ]]
```

In[∘]:=

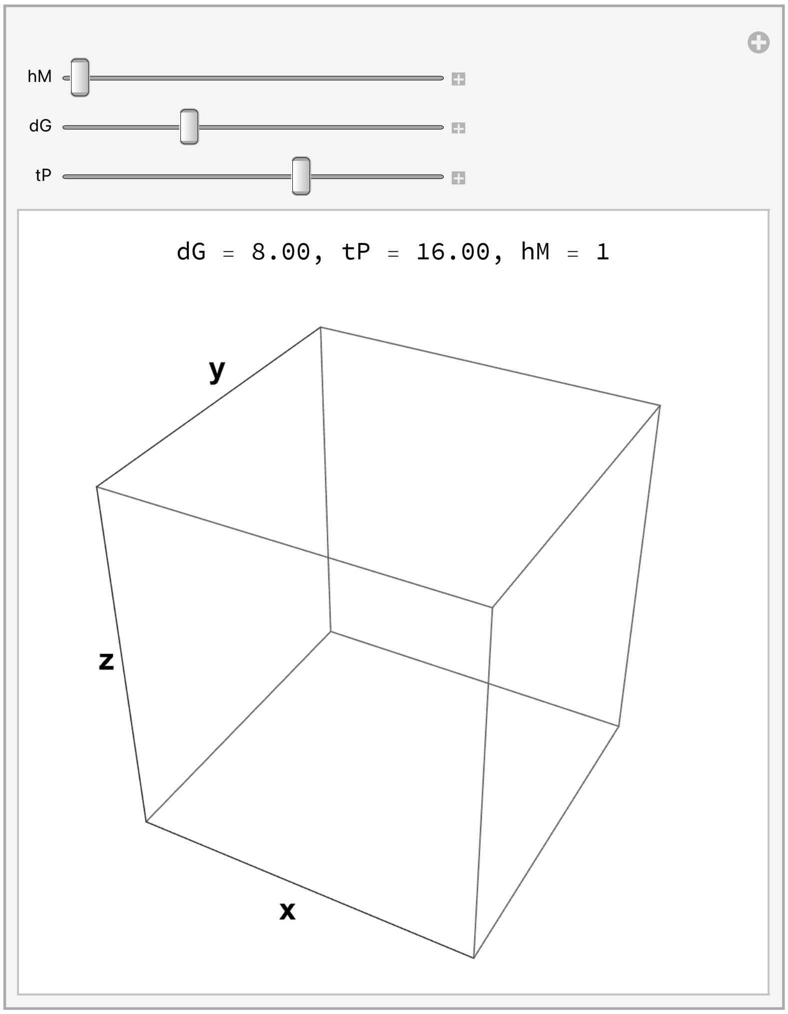
```
Manipulate[Labeled[Plot[Z3dfull[θ * π, dG, tP, hM, ϕ π], {θ, 0, 1}, PlotRange → All],
   Row[{"dG = ", NumberForm[dG, {3, 2}], ", tP = ", NumberForm[tP, {3, 2}],
     ", ϕ = ", NumberForm[ϕ, {3, 2}], ", hM = ", NumberForm[hM, {3, 2}]}], Top],
  {hM, 1, 2}, {dG, 0.1, 25}, {tP, 0.01, 25}, {ϕ, 0, 2}]
```

Out[∘]=



```
Manipulate[Labeled[
  SphericalPlot3D[Z3dfull[θ, dG, tP, hM, ϕ], {θ, 0, π}, {ϕ, 0, 2 π}, PlotRange → All,
   Axes → True, AxesLabel → {"x", "y", "z"}, Ticks → None, LabelStyle → {16, Bold}],
  Row[{"dG = ", NumberForm[dG, {3, 2}], ", tP = ", NumberForm[tP, {3, 2}], ", hM = ",
    NumberForm[hM, {3, 2}]}], Top], {hM, 1, 2}, {dG, 0.1, 25}, {tP, 0.01, 25}]
```

Out[ ]=



Just one simplify

Trig simplify function

$$\text{Sin}\left[2\,\text{ArcTan}\left[\frac{(-t+tP)\,\text{Sin}[\theta]\,\text{Sin}[\phi]}{dG+(t-tP)\,\text{Cos}[\theta]}\right]\right]$$

Simplifying $\text{Sin}\left[2\,\text{ArcTan}\left[\frac{(-t+tP)\,\text{Sin}[\theta]\,\text{Sin}[\phi]}{dG+(t-tP)\,\text{Cos}[\theta]}\right]\right]$

In[ ]:=
```
o = (-t + tP) Sin[θ] Sin[ϕ];
a = dG + (t - tP) Cos[θ];
```

In[ ]:=
```
hyp = Sqrt[a^2 + o^2]
```

Out[ ]=

$$\sqrt{(dG+(t-tP)\,\text{Cos}[\theta])^2+(-t+tP)^2\,\text{Sin}[\theta]^2\,\text{Sin}[\phi]^2}$$

```
sin2arctan3d = FullSimplify[2 * o/hyp * a/hyp] (*Sin(2x) = 2CosxSinx*)
 (*I graphed this against the original Sin[2ArcTan[]] and they look identical*)
```

Out[ ]=

$$\frac{2\,(-t+tP)\,(dG+(t-tP)\,\text{Cos}[\theta])\,\text{Sin}[\theta]\,\text{Sin}[\phi]}{(dG+(t-tP)\,\text{Cos}[\theta])^2+(t-tP)^2\,\text{Sin}[\theta]^2\,\text{Sin}[\phi]^2}$$

In[ ]:=
```
TrigSimp[(-t + tP) Sin[θ] Sin[ϕ], dG + (t - tP) Cos[θ]]
```

Out[ ]=

$$\frac{2\,(-t+tP)\,(dG+(t-tP)\,\text{Cos}[\theta])\,\text{Sin}[\theta]\,\text{Sin}[\phi]}{(dG+(t-tP)\,\text{Cos}[\theta])^2+(t-tP)^2\,\text{Sin}[\theta]^2\,\text{Sin}[\phi]^2}$$

Simplifying $\text{Sin}\left[2\,\text{ArcTan}\left[\frac{(-t+tP)\,\text{Cos}[\phi]\,\text{Sin}[\theta]}{dG+(t-tP)\,\text{Cos}[\theta]}\right]\right]$

In[ ]:=
```
o2 = (-t + tP) Cos[ϕ] Sin[θ];
```

In[∘]:=
```
hyp2 = Sqrt[a^2 + o2^2]
```

Out[∘]=

$$\sqrt{(dG + (t - tP)\,Cos[\theta])^2 + (-t + tP)^2\,Cos[\phi]^2\,Sin[\theta]^2}$$

```
sin2arctan3dalt = FullSimplify[2 * o2/hyp2 * a/hyp2]
 (*I graphed this against the original Sin[2ArcTan[]] and they look identical*)
```

Out[∘]=

$$\frac{2\,(-t + tP)\,(dG + (t - tP)\,Cos[\theta])\,Cos[\phi]\,Sin[\theta]}{(dG + (t - tP)\,Cos[\theta])^2 + (t - tP)^2\,Cos[\phi]^2\,Sin[\theta]^2}$$

3D metric, anisotropic wave source

Incorrect Plus Polarized Redshift in 3D (same as 2D)

Cross Polarization Redshift in 2D and incorrectly in 3D

(Updated approach) Redshift of + polarized rotated metric in 2D

Deflection angle in 2D